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SEVERO OCHOA SUMMER SCHOOL IN

MATERIALS FOR ENERGY



SUPERCONDUCTIVITY (Tutorial)



UNIVERSITÀ
DEGLI STUDI
DI GENOVA

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Outline

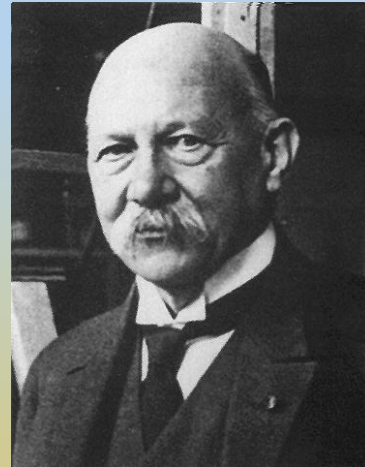
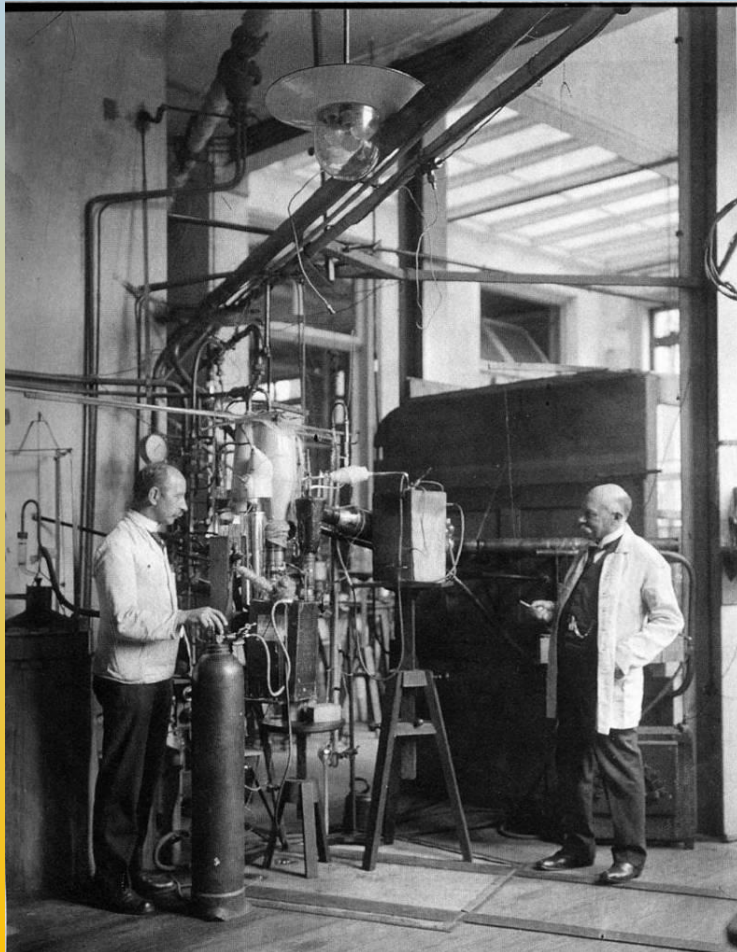
- ⇒ **Story of superconductivity phenomenon** going through the discovery of its main properties.
- ⇒ **Microscopic theory of superconductivity** and main parameters which characterize the superconducting state.
- ⇒ **Magnetic properties** and main mechanisms which allow a superconductors to carry superconducting current.
- ⇒ Main **superconducting materials**.
- ⇒ Peculiarities and issues of **novel superconducting materials**.

Kammerlingh Onnes, Leiden

1911: discovery of superconductivity

1908: Liquefaction of Helium

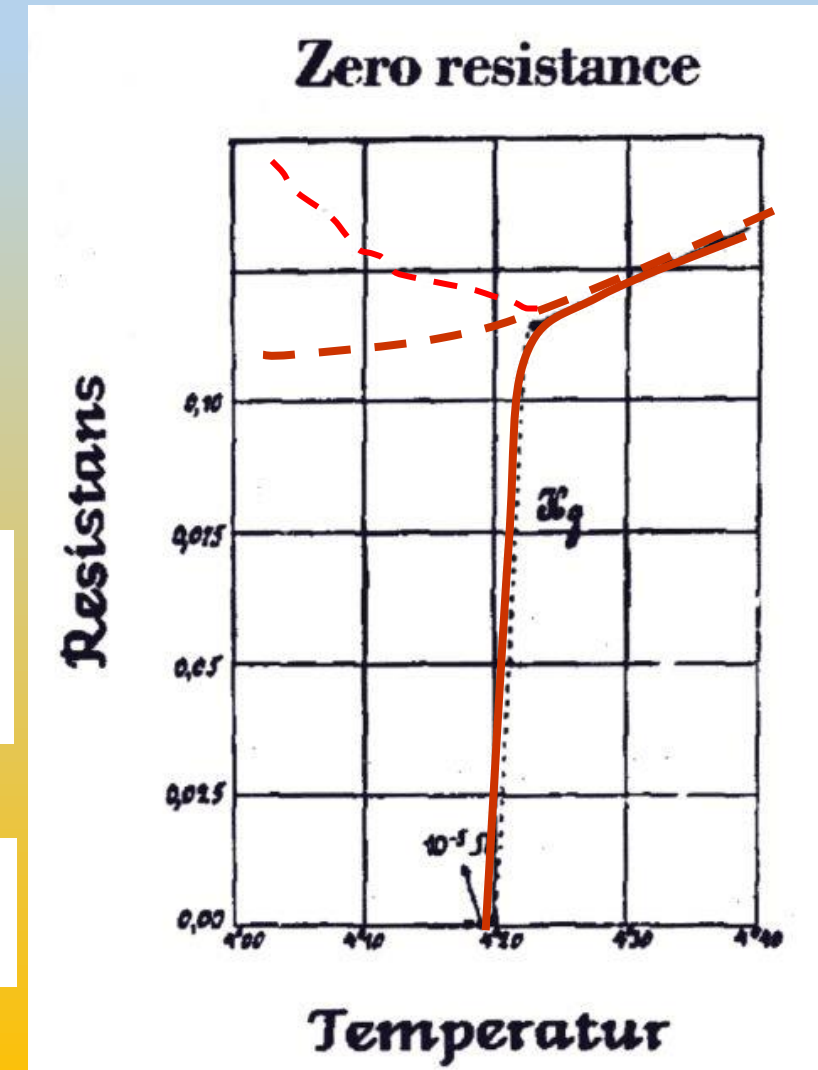
1913: Nobel price



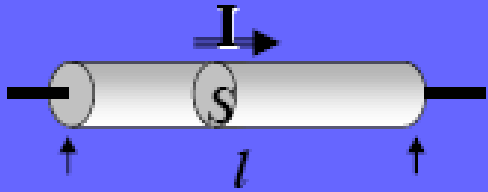
1913

⇒ He was measuring the resistivity of Hg to investigate the behavior of resistivity at low temperature

⇒ He noticed that the electrical resistance dropped to zero at 4.2K



The consequences of the Zero Resistance State ($R = 0$)



$$V = RI$$

$$R = \rho \frac{l}{S}$$

$R = 0$ implies no Joule dissipation in cable conducting current

$R = 0$ implies that current can flow in a loop without decay

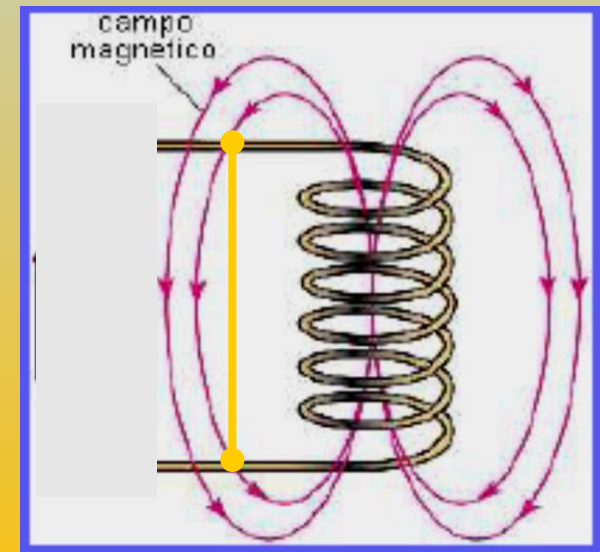
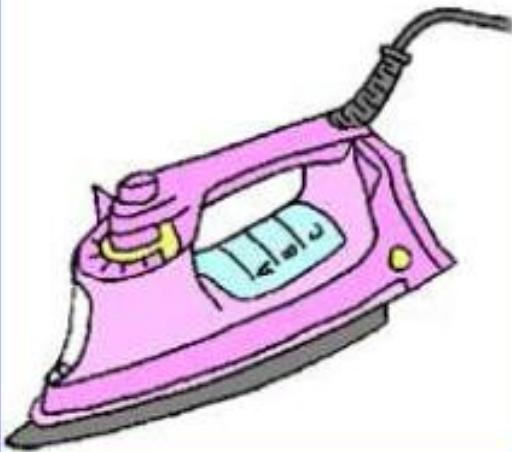
$R = 0$ implies that large magnetic field can be realized through solenoids with large current flowing in persistent mode



$$I = I_0 e^{-\frac{t}{\tau}}$$

$$\tau = \frac{L}{R}$$

$$P = RI^2$$



Kammerlingh Onnes's Visions

Construction of a 10 T Magnet with Hg und Pb Wires



Presented at 3rd International Congress on Refrigeration , Chicago 1913

Superconducting Magnet Coil

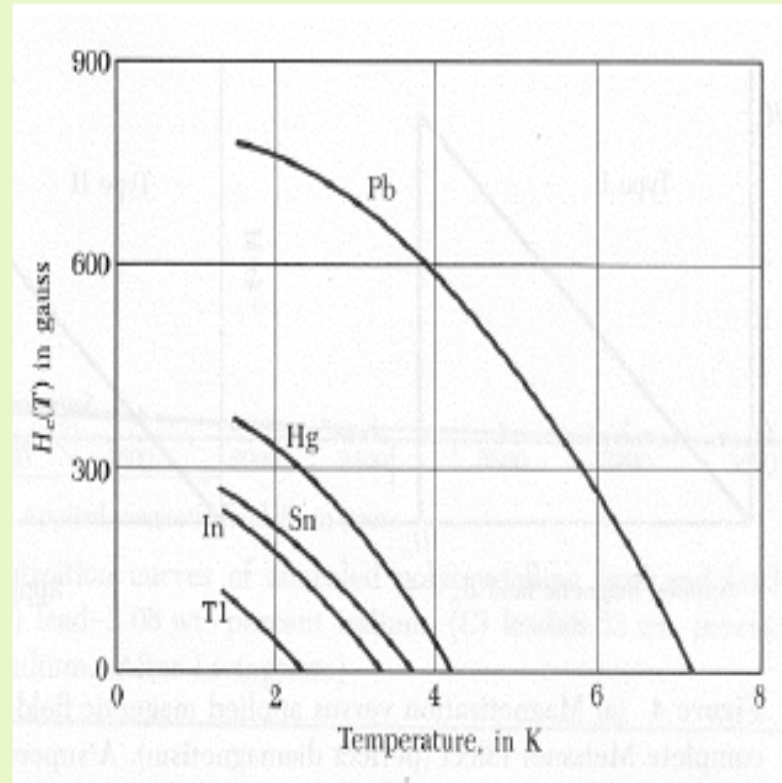
Experiments with Hg-and Pb wires failed

The coil lost superconducting properties already at small Current densities and at Magnetic Fields of several 100 Gauss.

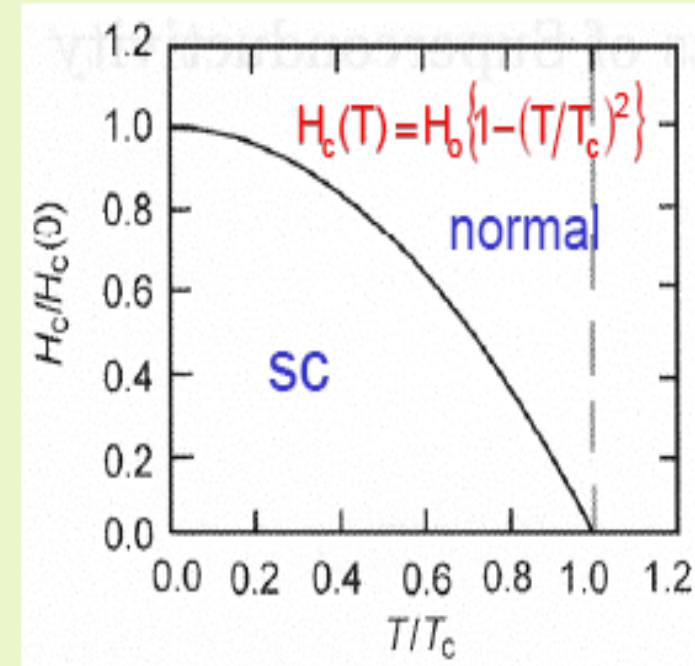
Superconductivity can be destroyed also by an external magnetic field , which is called **critical field, H_c** (Kammerlingh Onnes 1914)

Critical Field

Temperature dependence of $H_c(T)$



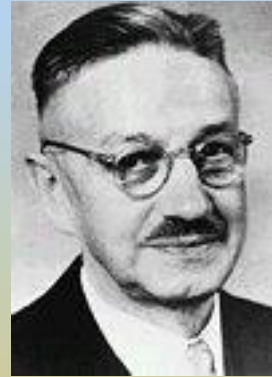
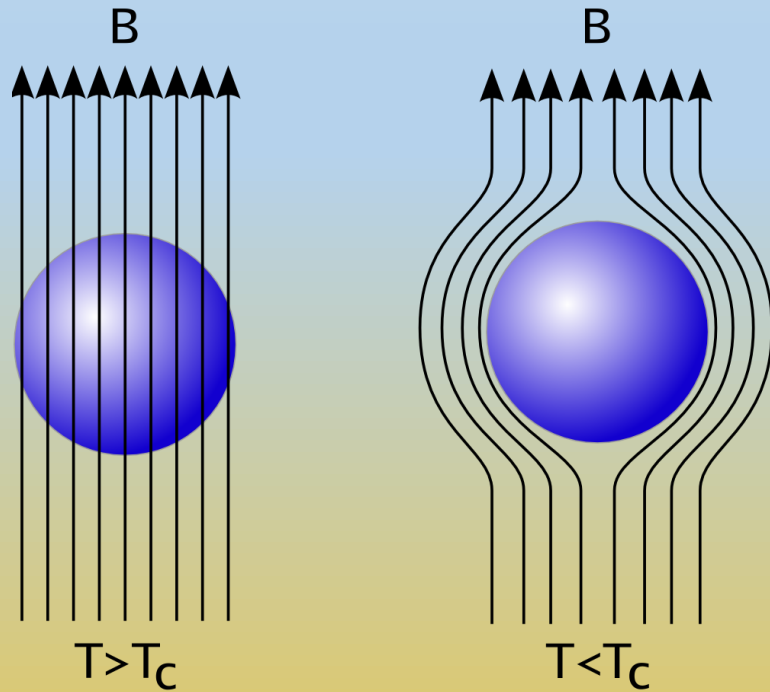
All curves can be collapsed onto a similar curve after re-scaling



Empirically :

$$H_c(T) = H_c(0) \left[1 - (T/T_c)^2 \right]$$

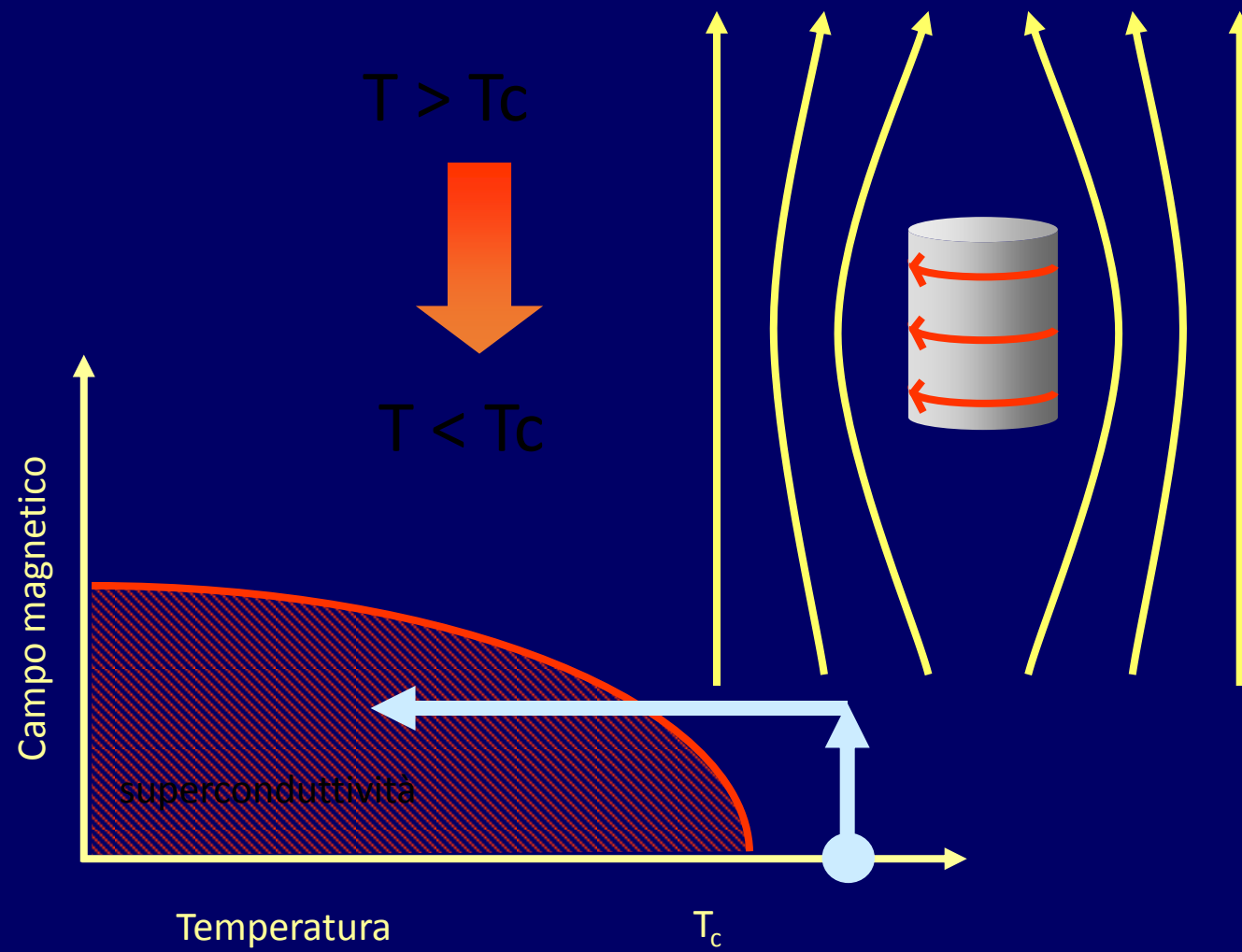
1933: Meissner-Ochsenfeld effect



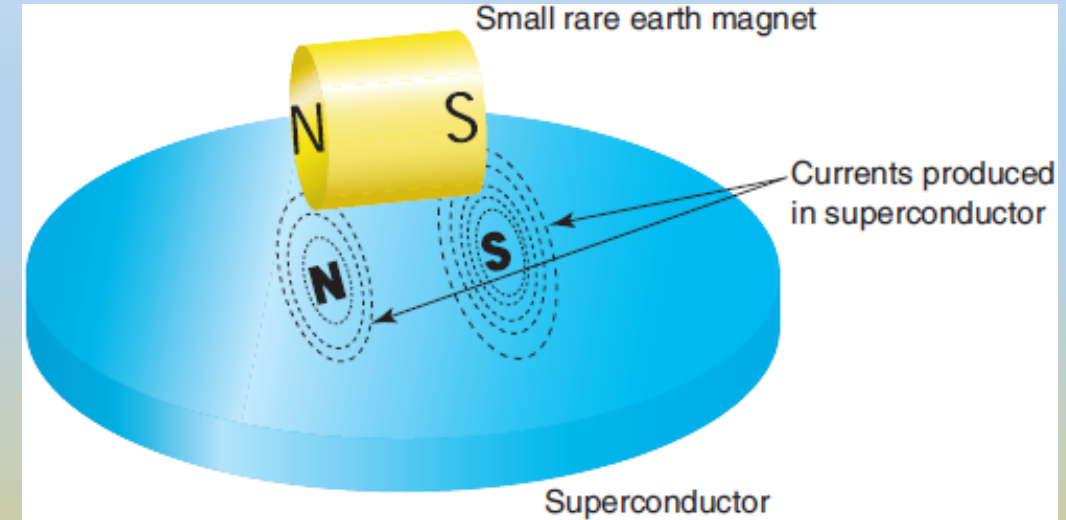
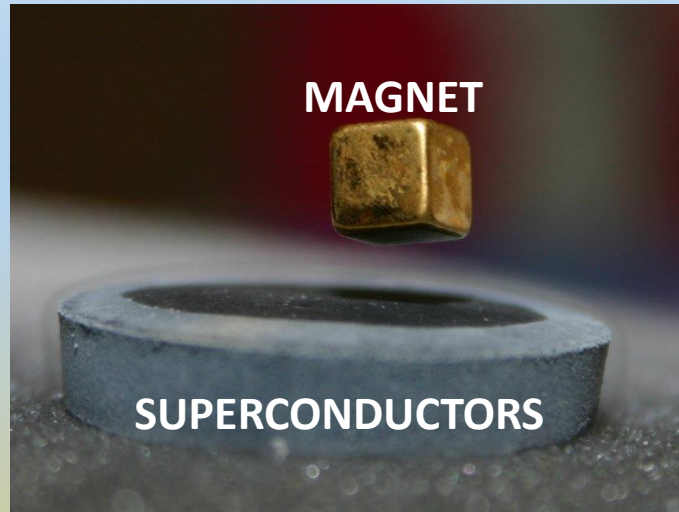
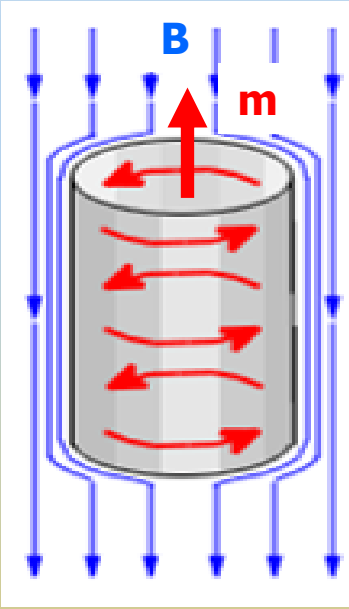
Magnetic field does not penetrate the superconductor

Ideal conductor!

Ideal diamagnetic!



The consequences of the perfect diamagnetisms $\chi = -1$



The expulsion of the magnetic field, generates **great magnetic moments**, able to levitate extremely large masses like the coaches of a train



ELECTRODINAMIC LEVITATION

The future (2025) maglev train between **Tokyo and Osaka** will be the **fastest train** in the world with a speed approximately **1000 km/h.**

The quantum coherence

1950: Ginzburg-Landau Phenomenology Ψ -Theory of Superconductivity

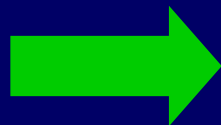
$T > T_c$



$T < T_c$



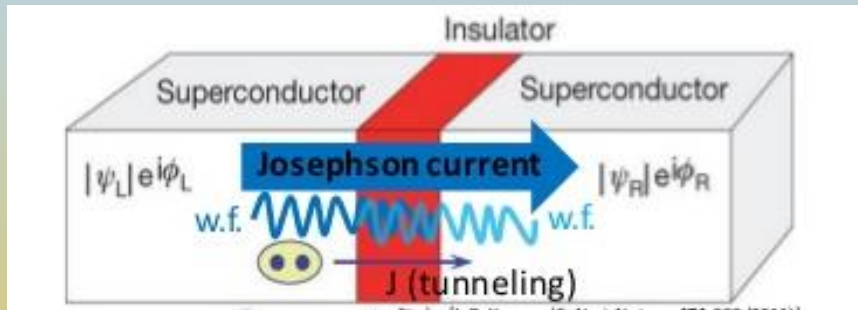
A macroscopic wave function describes the system as a whole



Macroscopic quantum effects

1962: Josephson effect

Super electrons cross the insulating junction between the two superconductors as a consequence of the phase difference between the two superconductors

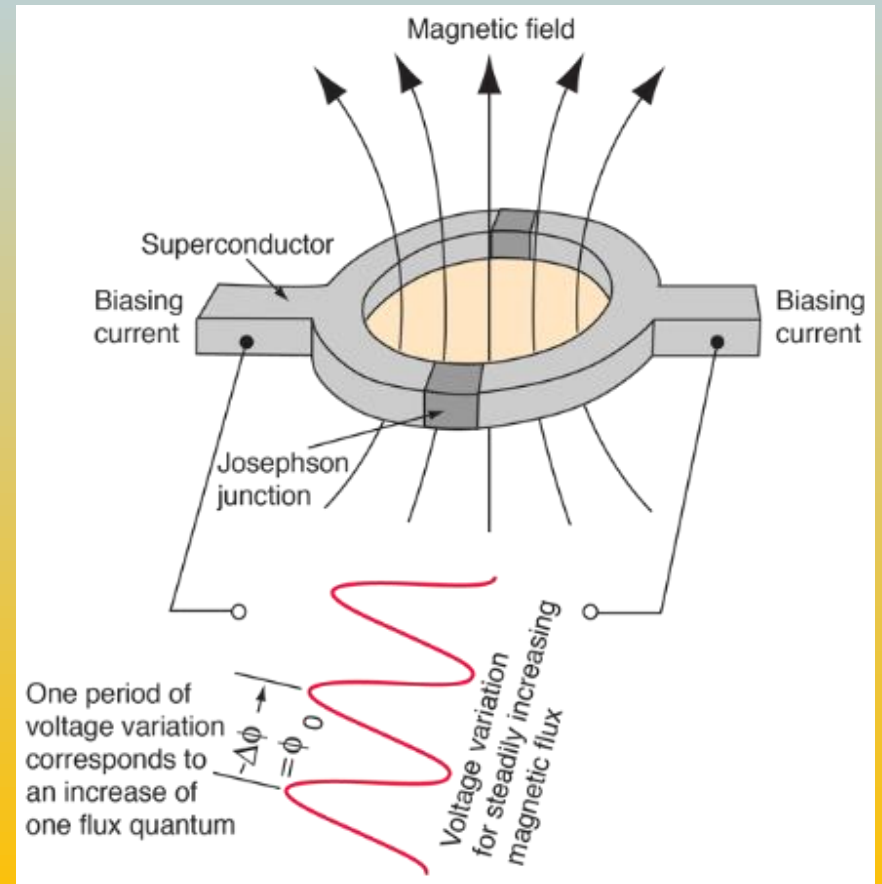
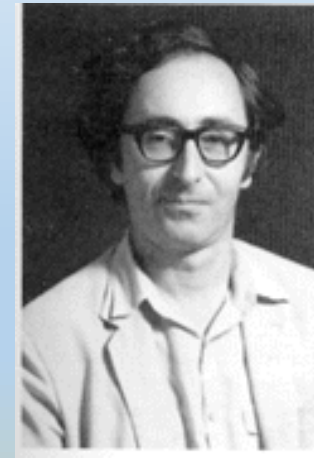


SQUID superconducting quantum interference device

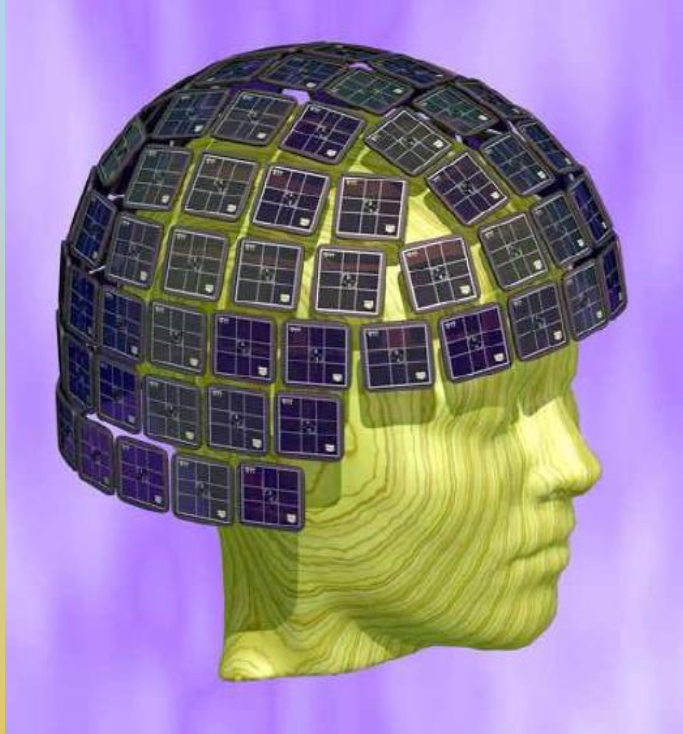
it is based on the presence of two superconductive junctions and on the interference due to the phase difference across the two junctions



1973



SQUID for biomagnetisms



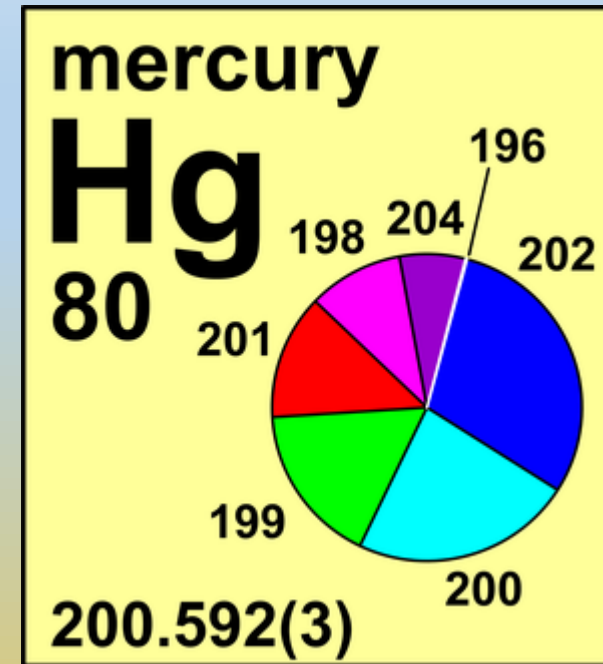
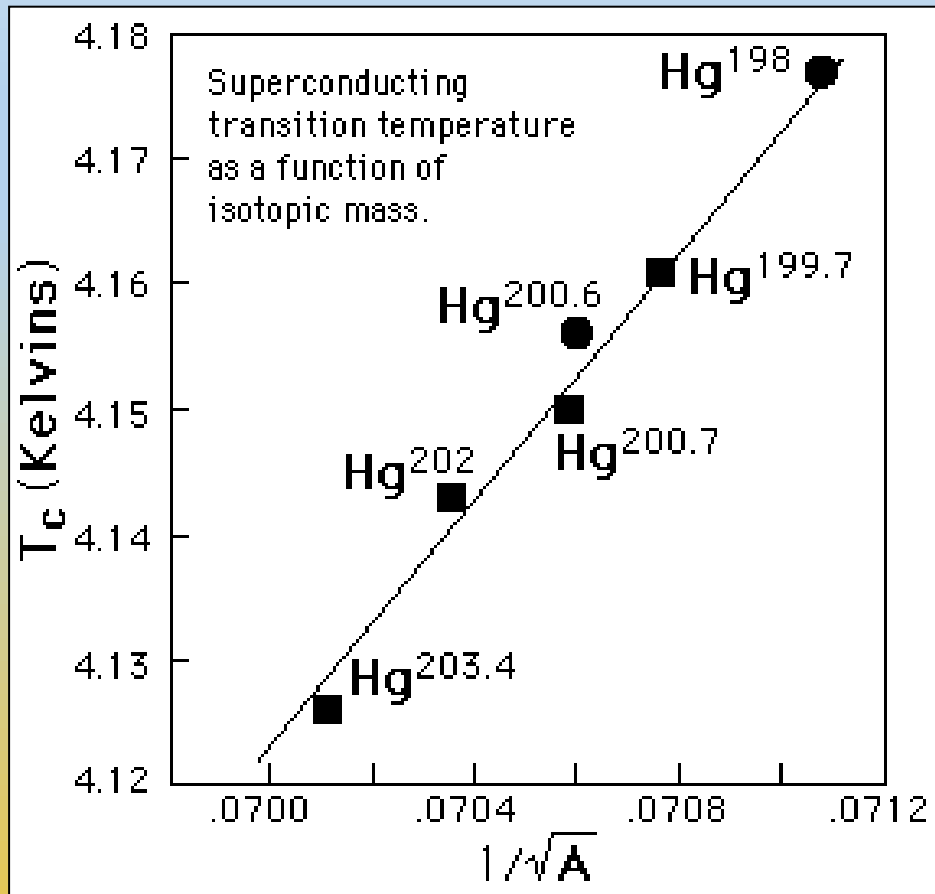
SQUID sensitivity	10^{-14} T
Heart magn. field	10^{-10} T
Brain magn. field	10^{-13} T
Earth magn. field	10^{-5} T
Fridge magnet	10^{-2} T

1957: Microscopic theory of superconductivity (Bardeen, Cooper, Schrieffer)



1972

1950: Isotopic effect

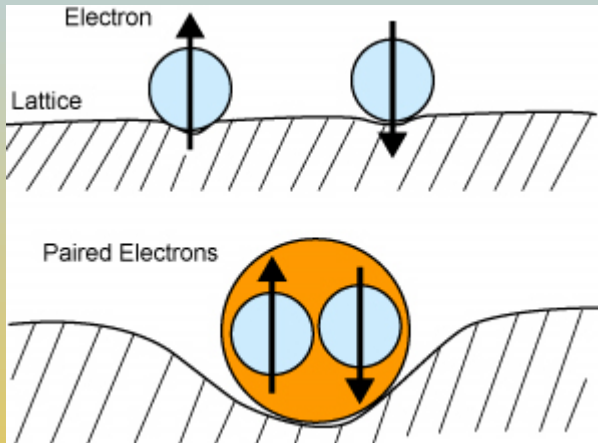


Isotopic effect suggests that the ion lattice plays a crucial role

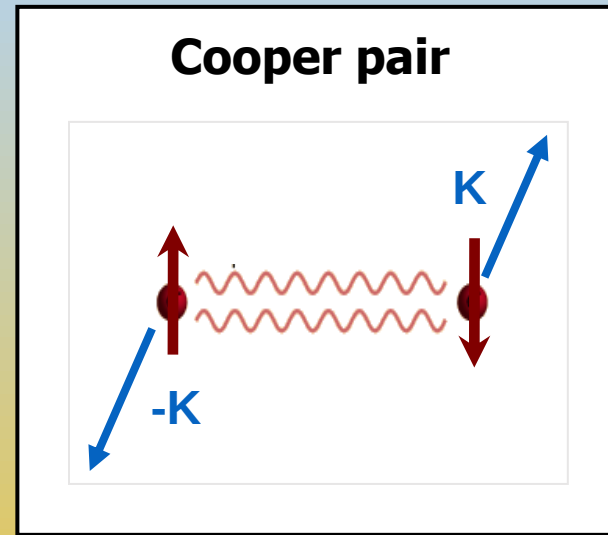
$$T_c \propto M^{-1/2}, \quad H_c \propto M^{-1/2}.$$

BCS- Theory of superconductivity

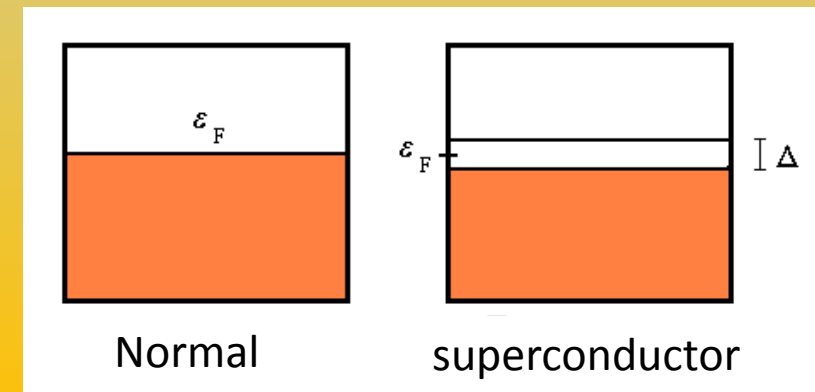
■ in the presence of an **attractive interaction** provided by the lattice



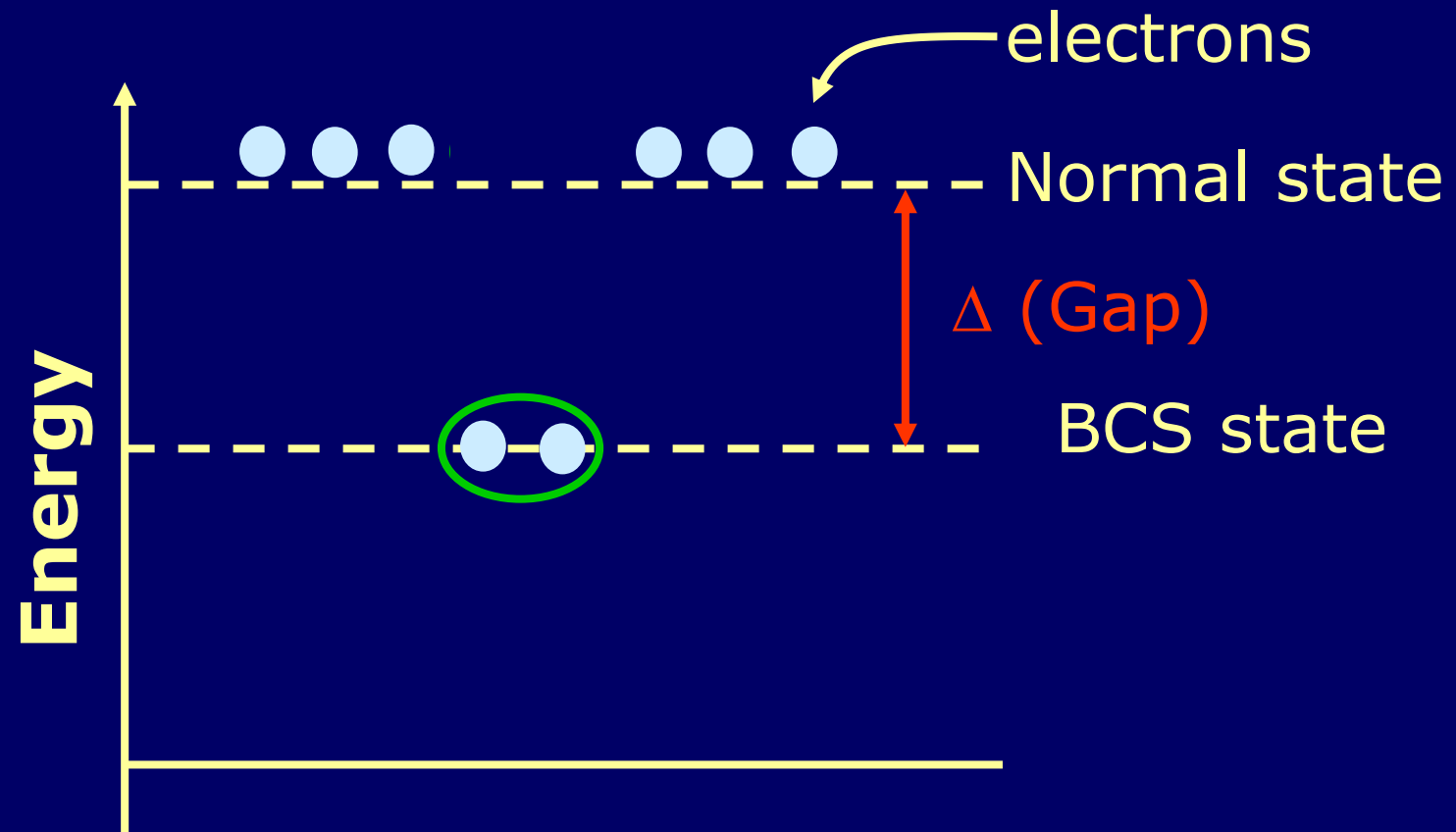
■ electrons form pairs of opposite impulse and spin: the **Cooper pairs**

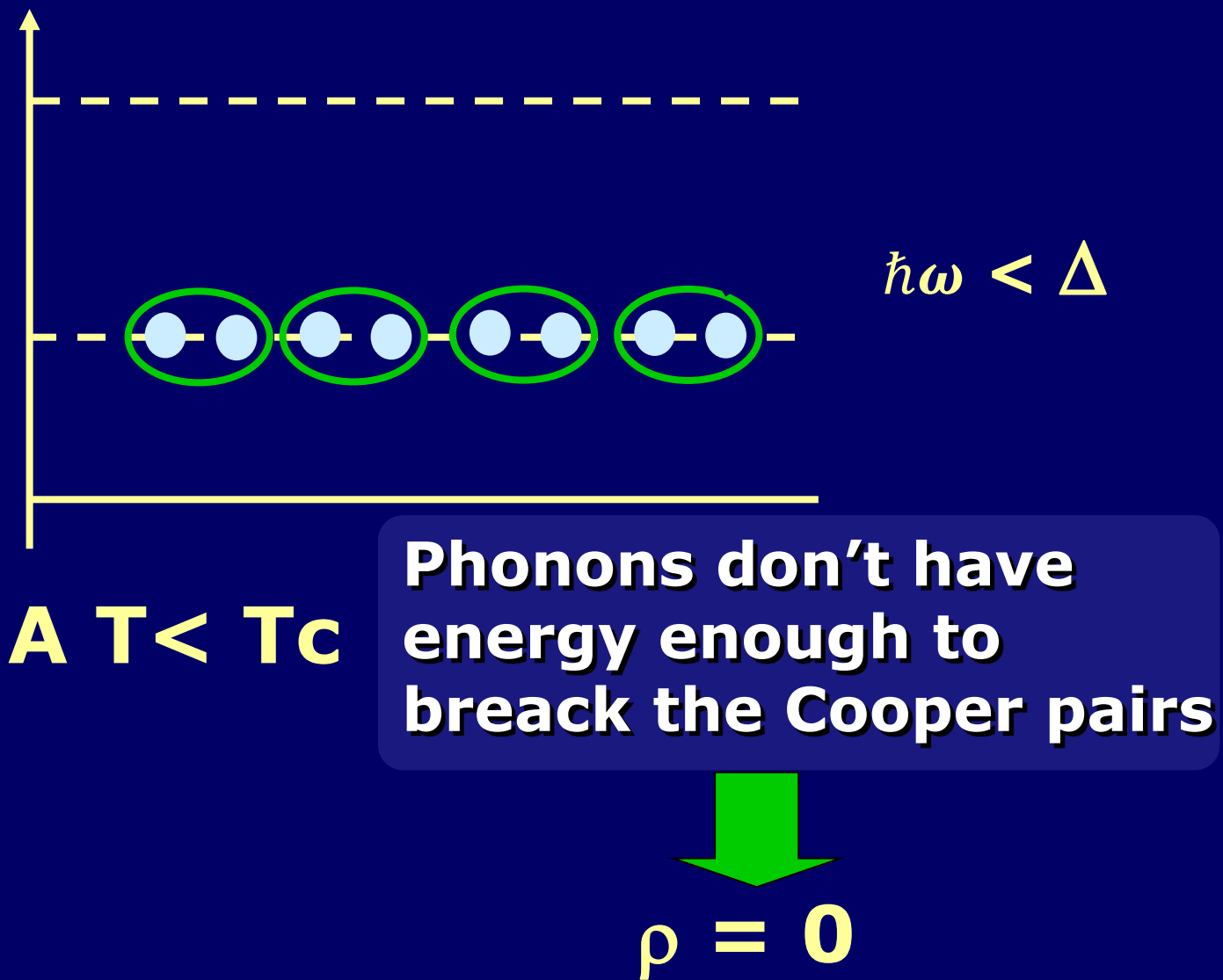


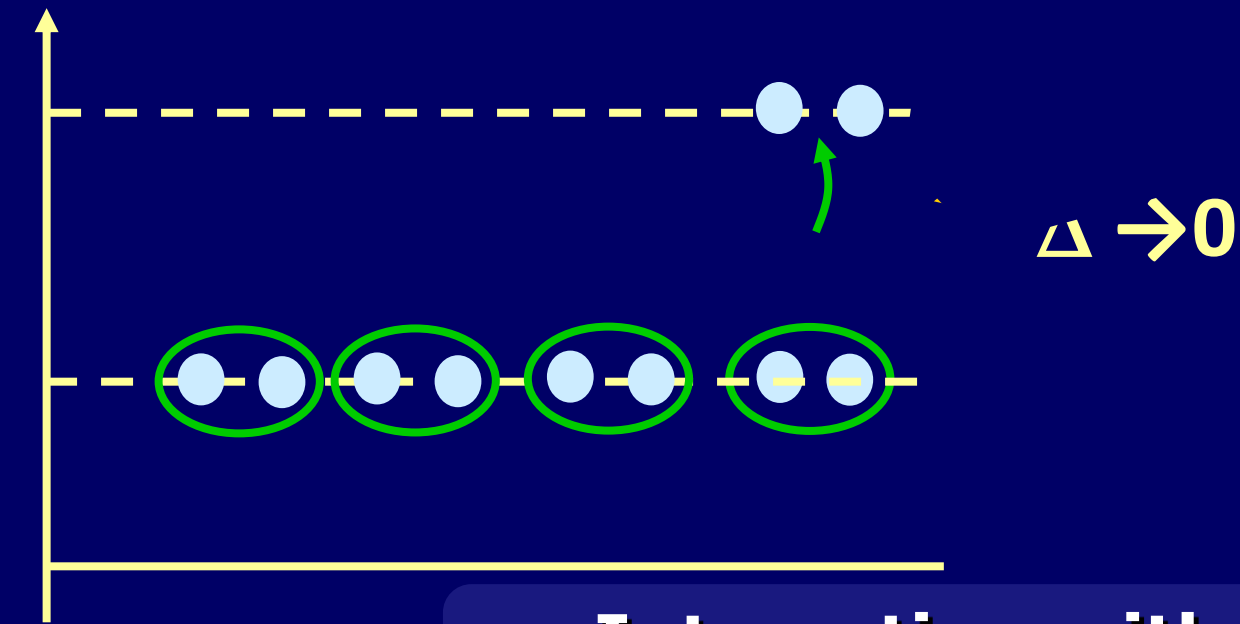
■ An **energy gap opens** in the energy spectrum which stabilizes the superconducting state



Energy gap



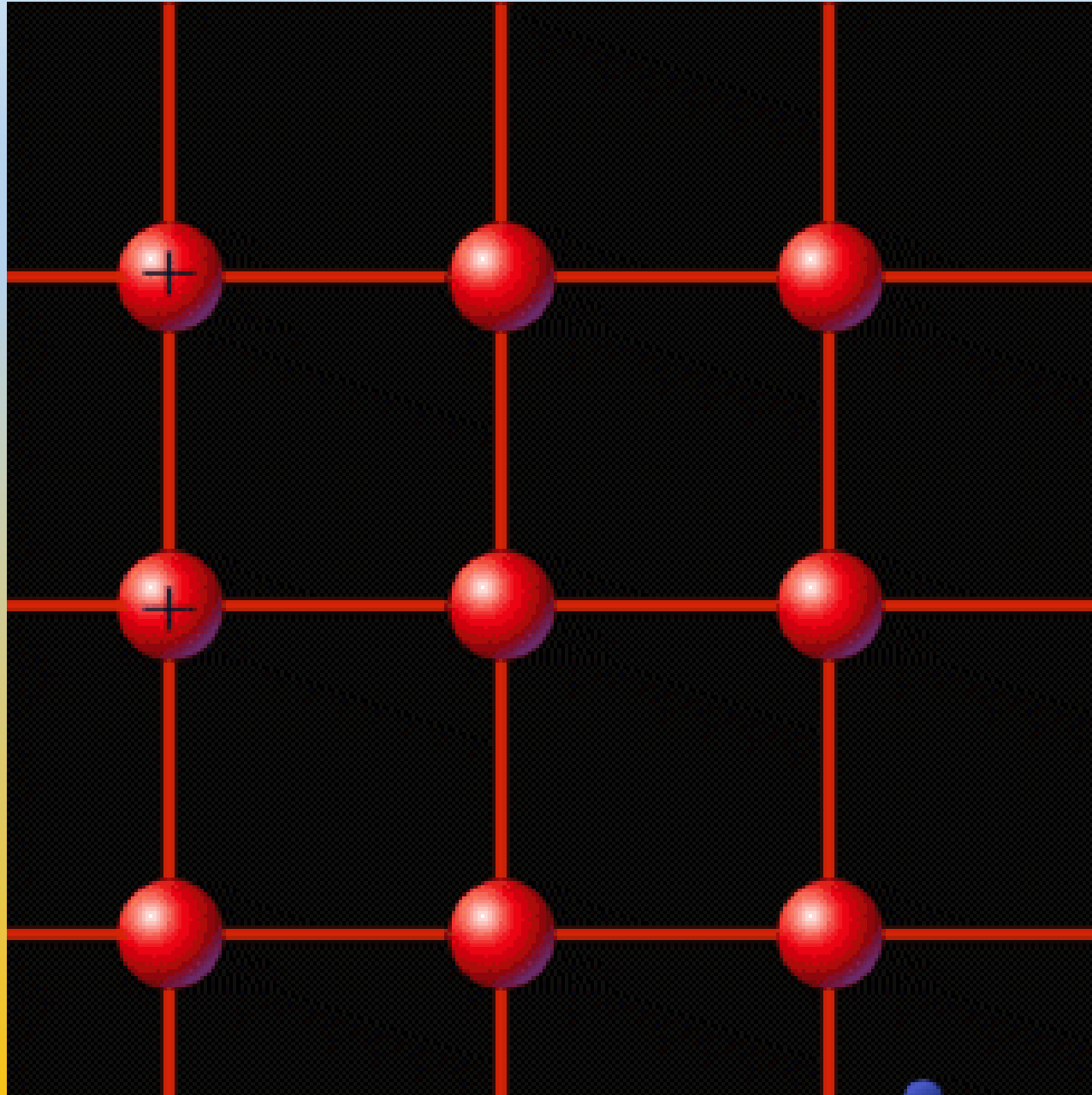




**Interaction with
phonons break the
Cooper pairs**

$\rho \neq 0$

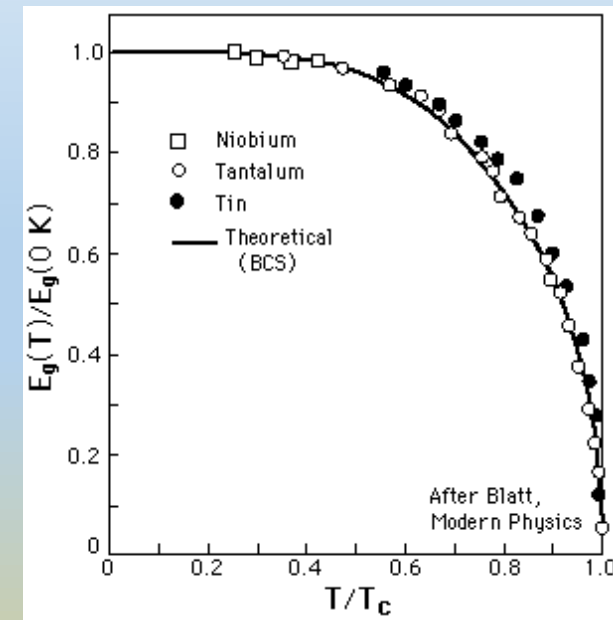
Electron pairing mechanisms



BCS theory predictions

Energy Gap

$$\Delta = \hbar\omega_D \exp\left(-\frac{1}{g}\right)$$



Critical temperature

$T_c \sim 1\text{-}20 \text{ K}$

$$T_c = \Theta_D \exp\left(-\frac{1}{g}\right)$$

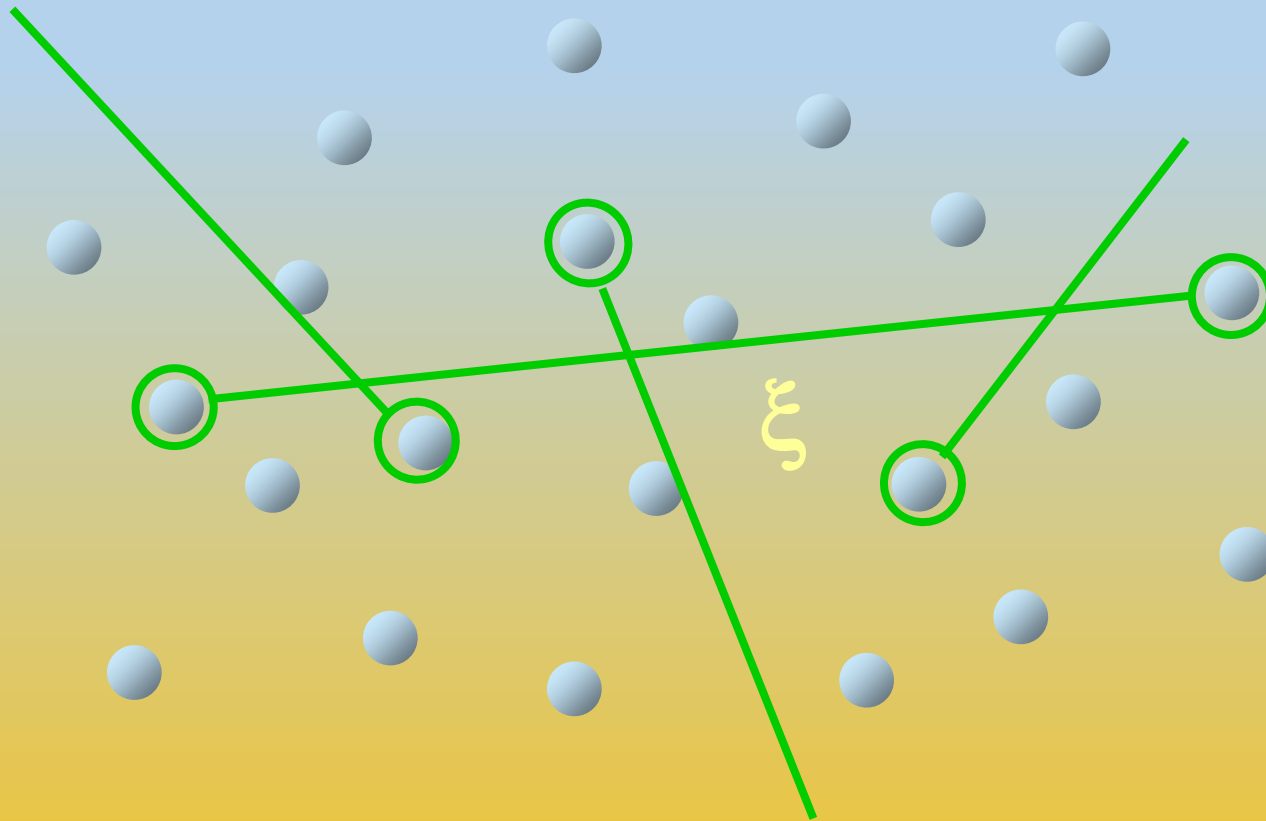
Θ_D Debye Temperature

$10^{-2} - 10^{-1}$

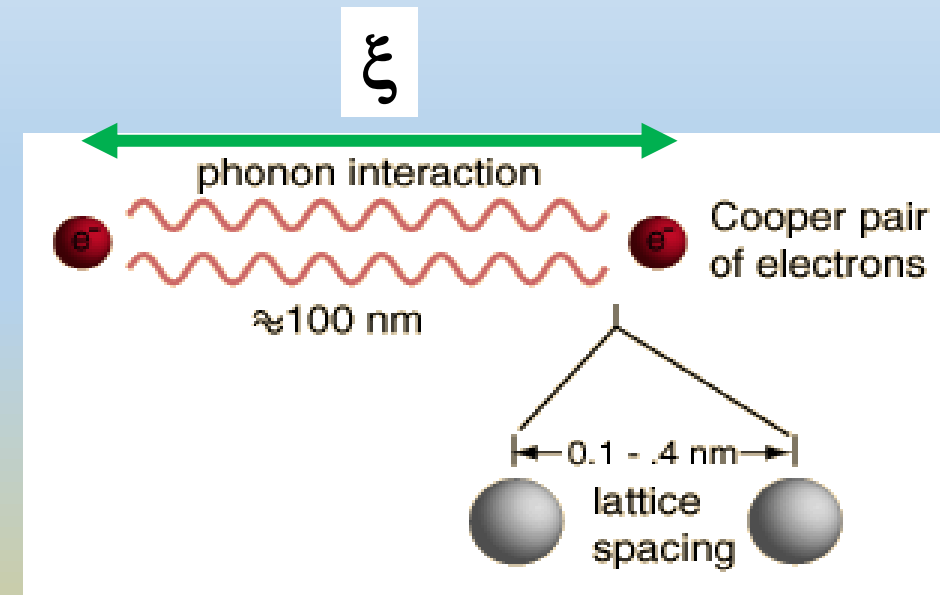
$T_{c\text{max}} \sim 30\text{-}40 \text{ K}$

$\Theta_D \sim 100\text{-}400 \text{ K}$

Coherence length



⇒ $\xi \sim$ size of the Cooper pair

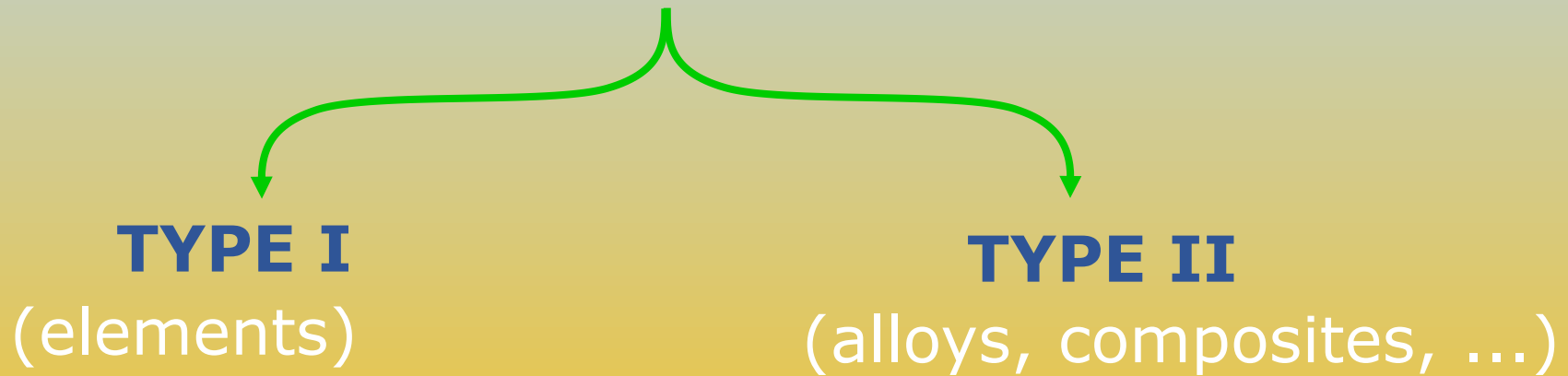


For the superconducting elements the coherence length can be as large as hundreds nm, much larger than the lattice spacing.

⇒ between the two electrons of a pair, there may be thousands of electrons belonging to other pairs.

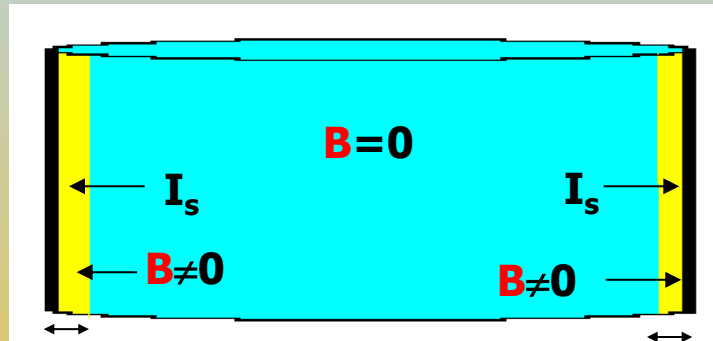
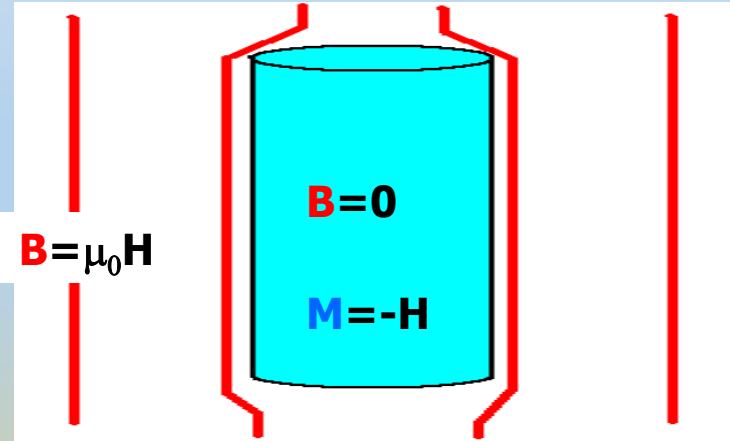
MAGNETIC PROPERTIES

**2 classes of superconductors
with different properties**

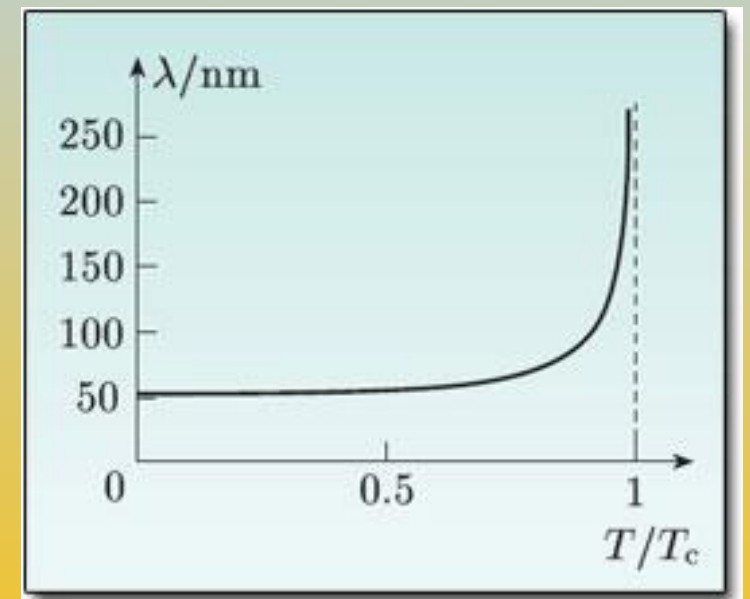
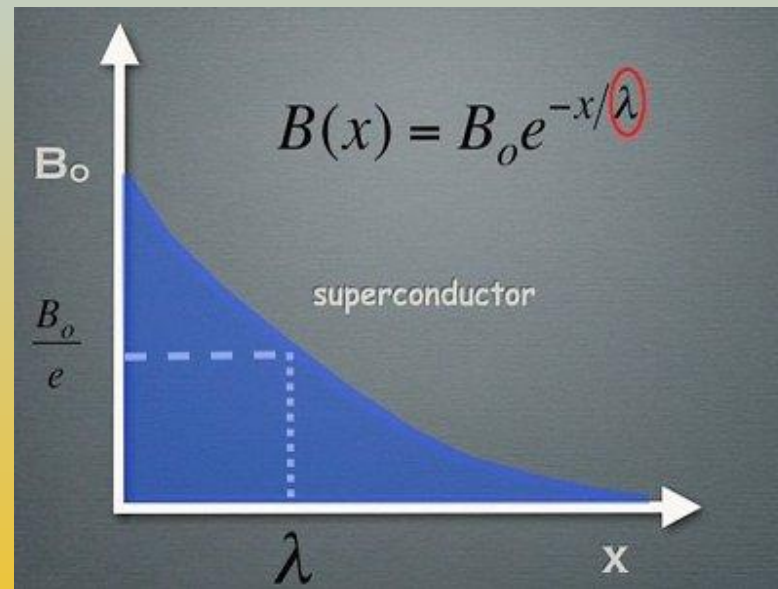


London penetration depth

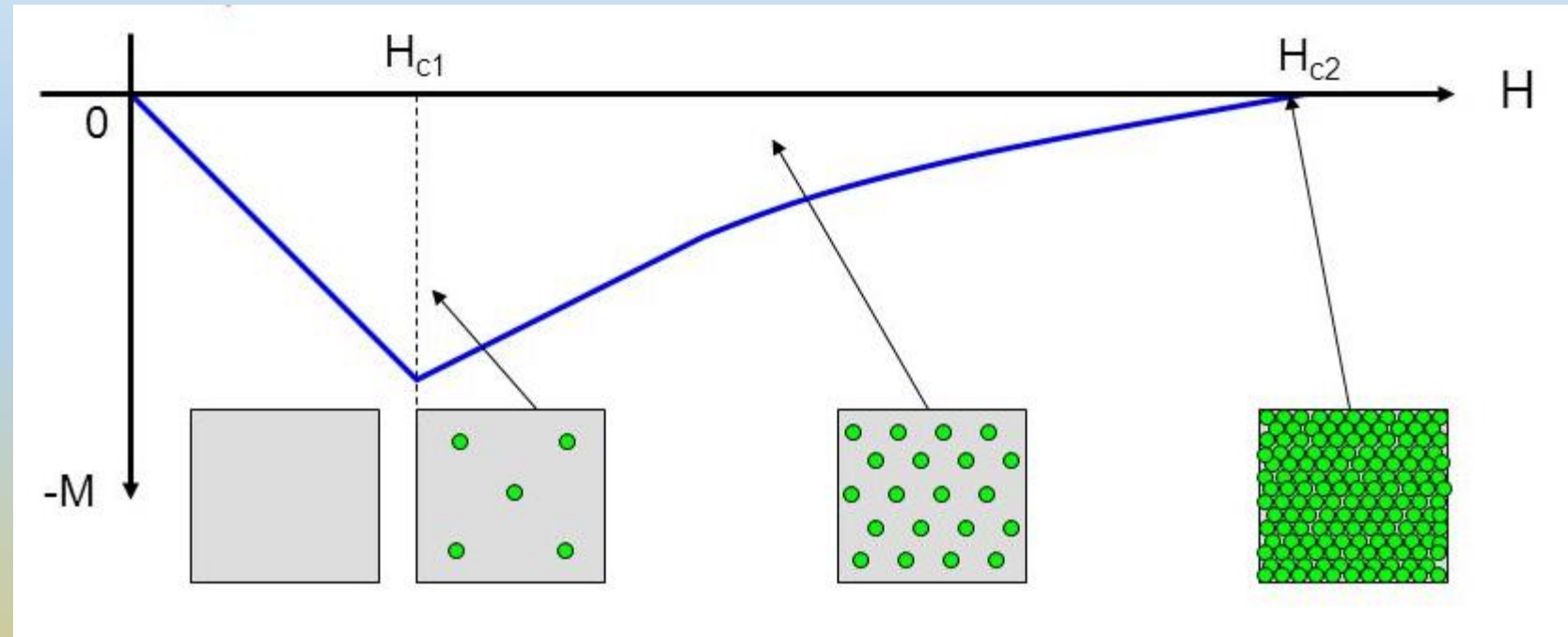
1935: Brothers London theory



The magnetic field does not change abruptly to zero within the superconducting material. In the superficial thickness where the superconducting currents flow, the magnetic field goes progressively to zero

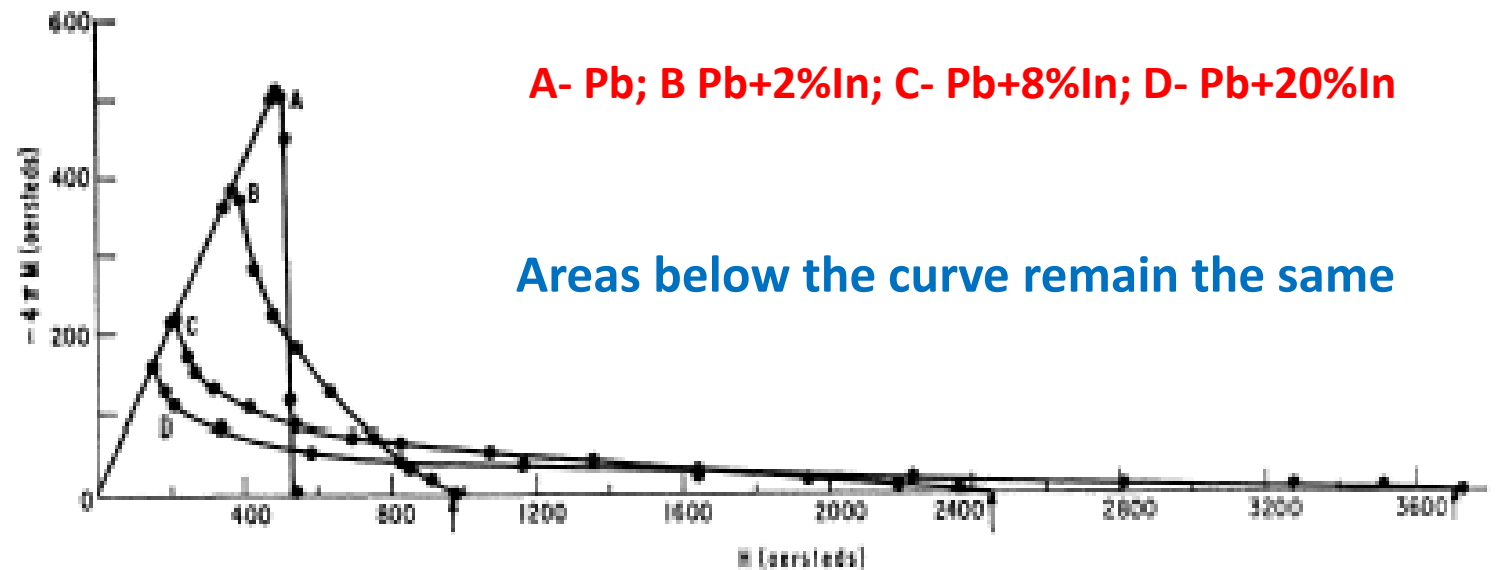


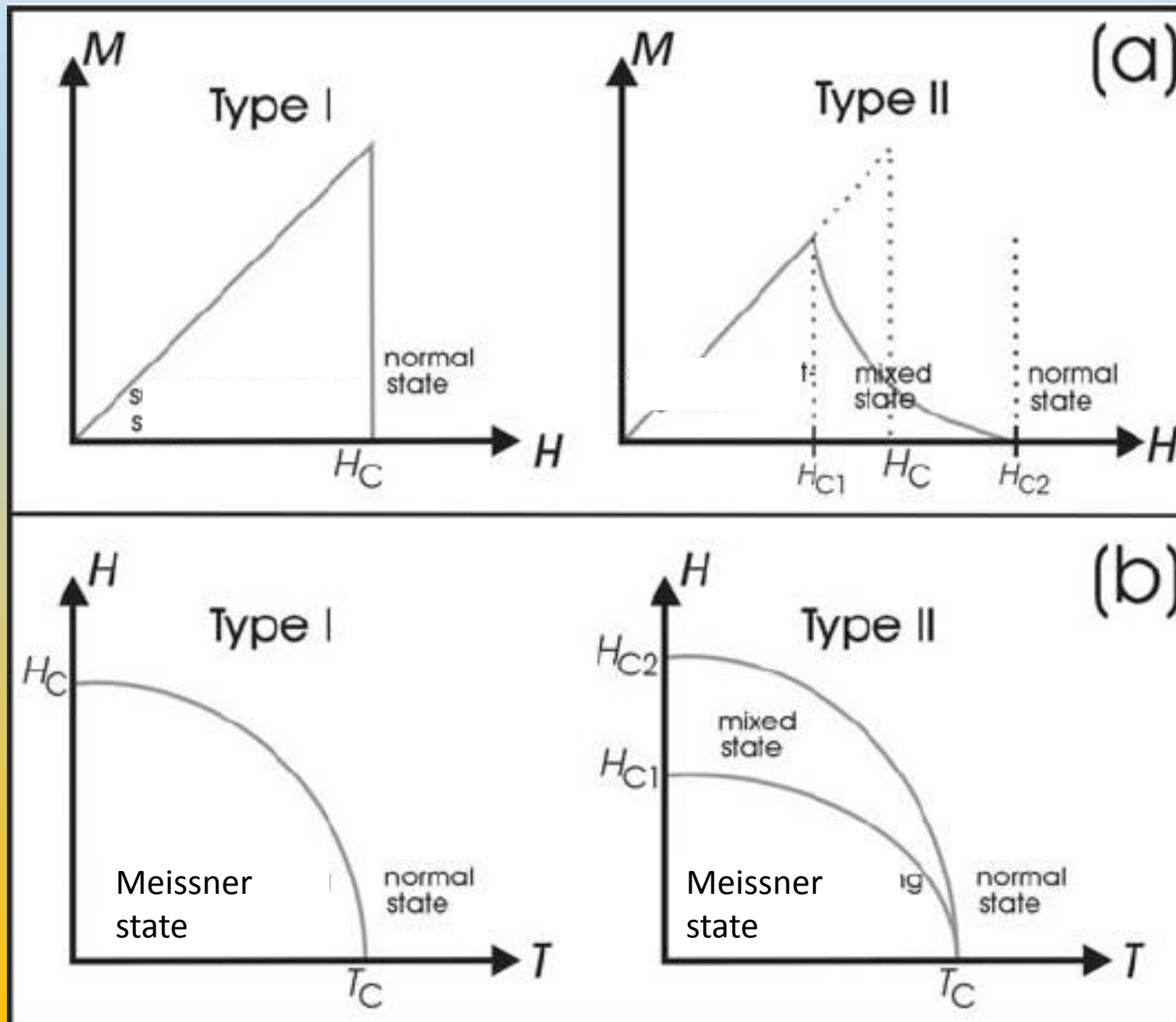
Magnetization curves



$$H_{c2} > H_c$$

Type II superconductors
can be high field
superconductors





Who is of I or II type?

$$\kappa = \frac{\lambda}{\xi}$$

Ginzburg - Landau parameter

$$\kappa < \frac{1}{\sqrt{2}} \quad \text{Type I} \quad ; \quad \kappa > \frac{1}{\sqrt{2}} \quad \text{Type II}$$

Type I

Material	T_c (K)	ξ (nm)	λ (nm)	κ (λ/ξ)
Cd	0.56	760	110	0.14
Al ^a	1.18	550	40	0.03
In ^a	3.41	360	40	0.11
Sn ^a	3.72	180	42	0.23
Ta	4.4	93	35	0.38
Pb ^a	7.20	82	39	0.48
Nb ^a	9.25	39	50	1.28
Pb-In	7.0	30	150	5.0
Pb-Bi	8.3	20	200	10
Nb-Ti	9.5	4	300	75
Nb-N	16	5	200	40
PbMo ₆ S ₈ (Chevrel)	15	2	200	100
V ₃ Ga (A15)	15	= 2.5	90	= 35
V ₃ Si (A15)	16	3	60	20
Nb ₃ Sn (A15)	18	3	65	22
Nb ₃ Ge (A15)	23.2	3	90	30
K ₃ C ₆₀	19	2.6	240	92
Rb ₃ C ₆₀	29.6	2.0	247	124
(La _{0.925} Sr _{0.075}) ₂ CuO ₄ ^b	37	2.0	200	100
YBa ₂ Cu ₃ O ₇ ^b	89	1.8	170	95
HgBaCaCuO	126	2.3		
HgBa ₂ Ca ₂ Cu ₃ O _{8+δ}	131			100
MgB ₂	39	10	80	10
SmFeAsO _{0.7} F _{0.3}	58	2.5	200	80
KBaFe2As2	38	3	200	67
Fe(Se,Te)	18	1.5	400	300

Type II

With disorder → Type II

$$\lambda = \left(\frac{m}{2e^2 \mu_0 n_s} \right)^{1/2}$$

$$\xi \propto \frac{n_s^{1/3}}{T_c}$$

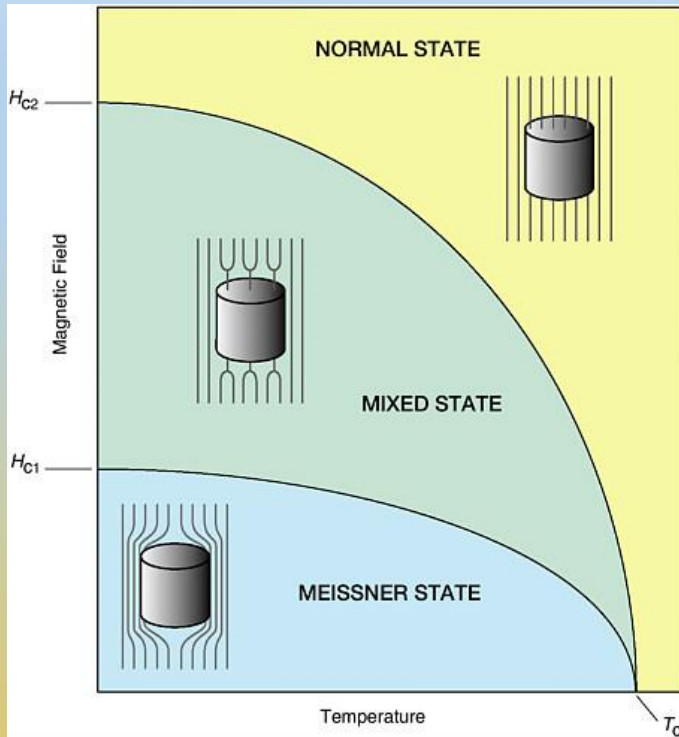
$$\kappa = \frac{\lambda}{\xi}$$

↑

↓

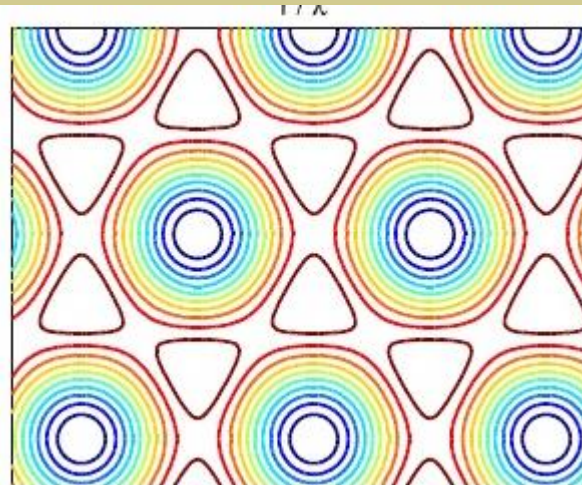
↑

1957: What the mixed state is (Abrikosov)

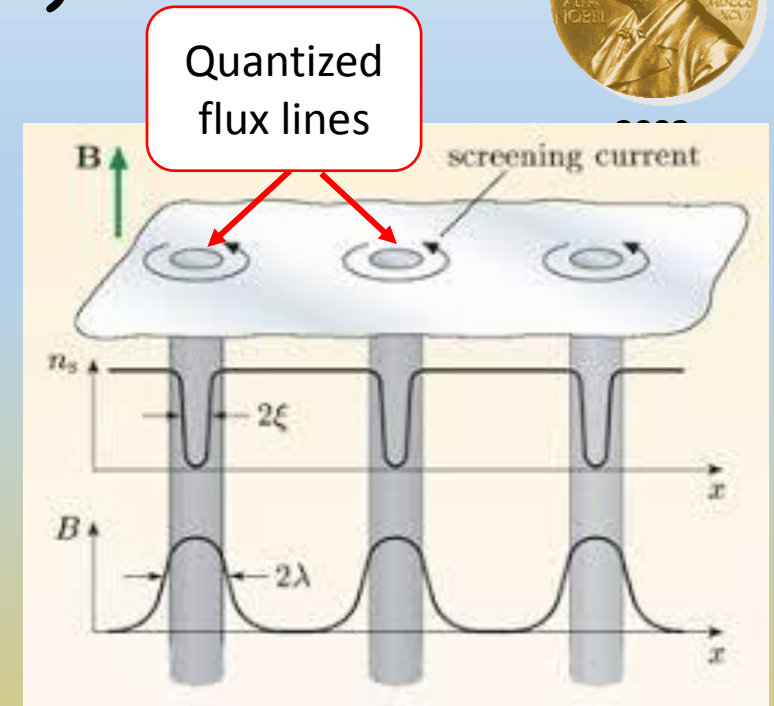


⇒ Quantized flux lines
(fluxons/vortices) enter the
superconductor

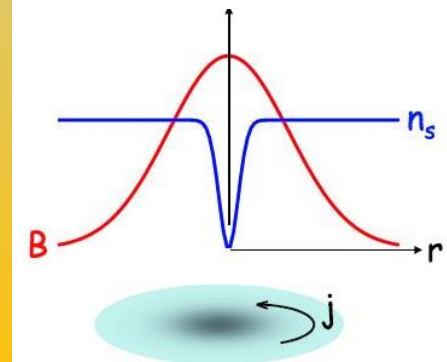
⇒ **Fluxons** form an exagonal lattice

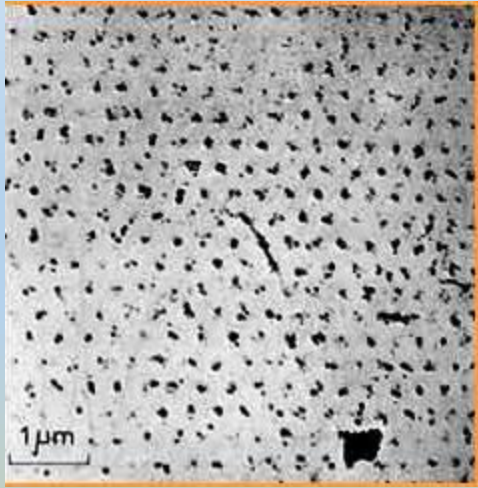


Abrikosov solution near H_{c2} –
Contours of B

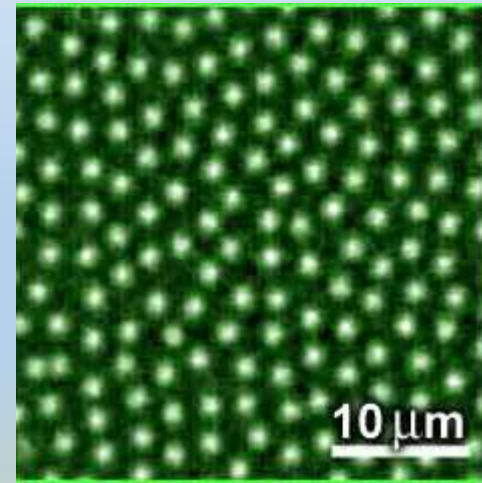


$$\Phi_0 = \frac{h}{2e} \\ 2.067 \times 10^{-15} \text{ Wb}$$

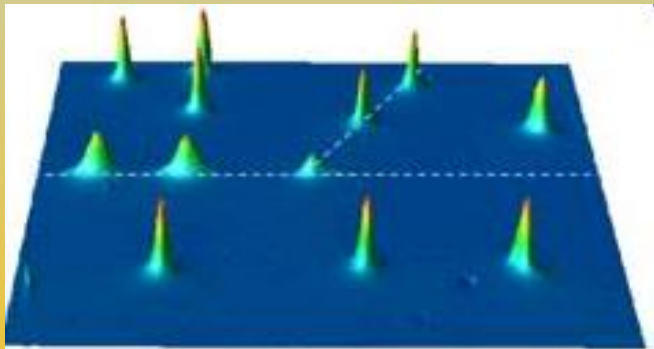




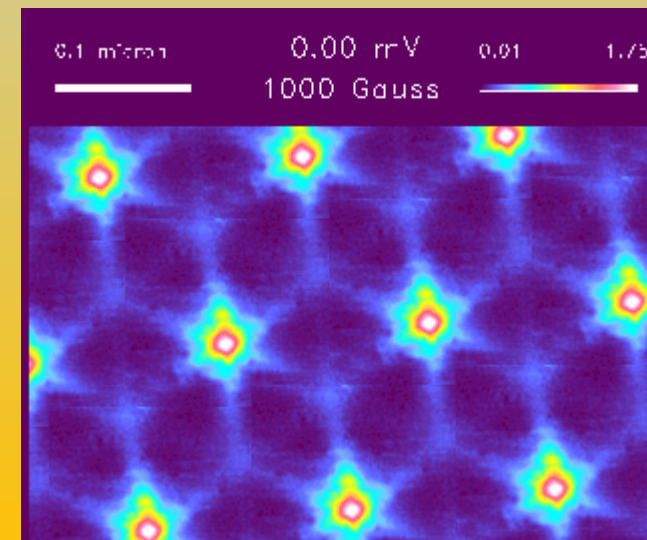
U. Essmann and H. Trauble
 Max-Planck Institute, Stuttgart
[Physics Letters 24A, 526 \(1967\)](#)



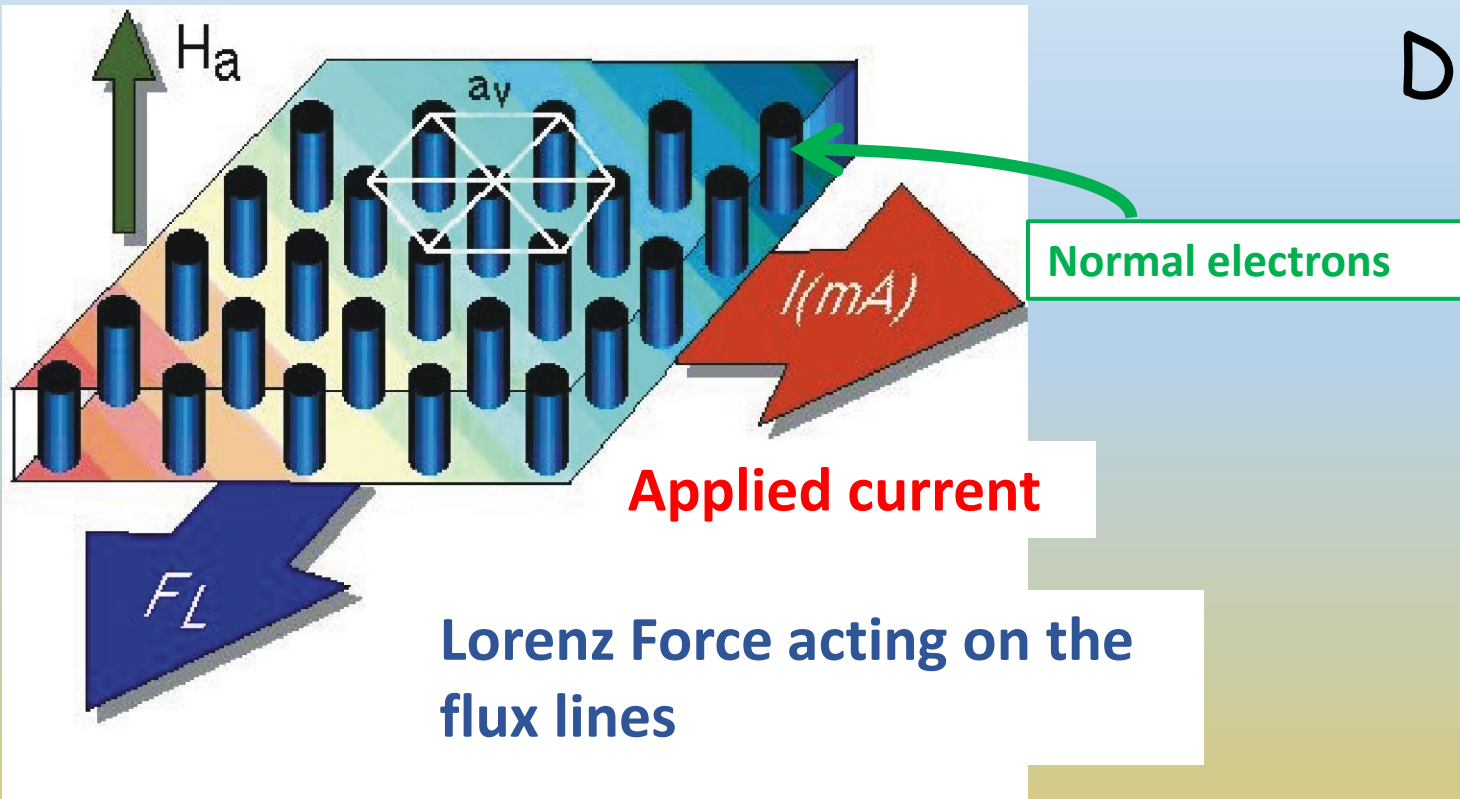
**Magneto-optical image
 of Vortex lattice, 2001**
 P.E. Goa et al.
 University of Oslo
[Supercond. Sci. Technol. 14, 729 \(2001\)](#)



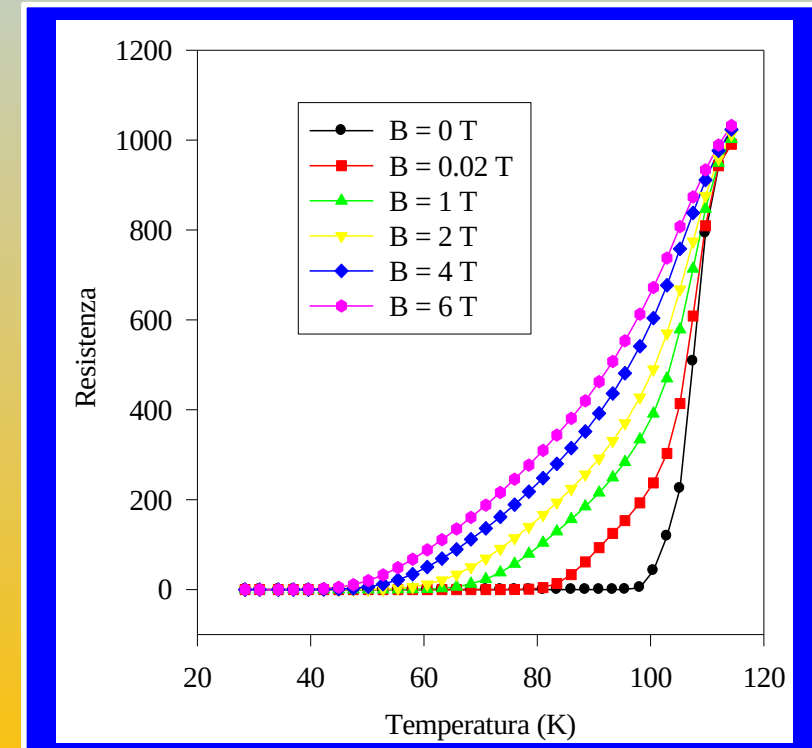
Scanning SQUID Microscopy of half-integer vortex, 1996
 J. R. Kirtley et al. [IBM Thomas J. Watson Research Center](#)
[Phys. Rev. Lett. 76, 1336 \(1996\)](#)



Dissipation in the mixed state



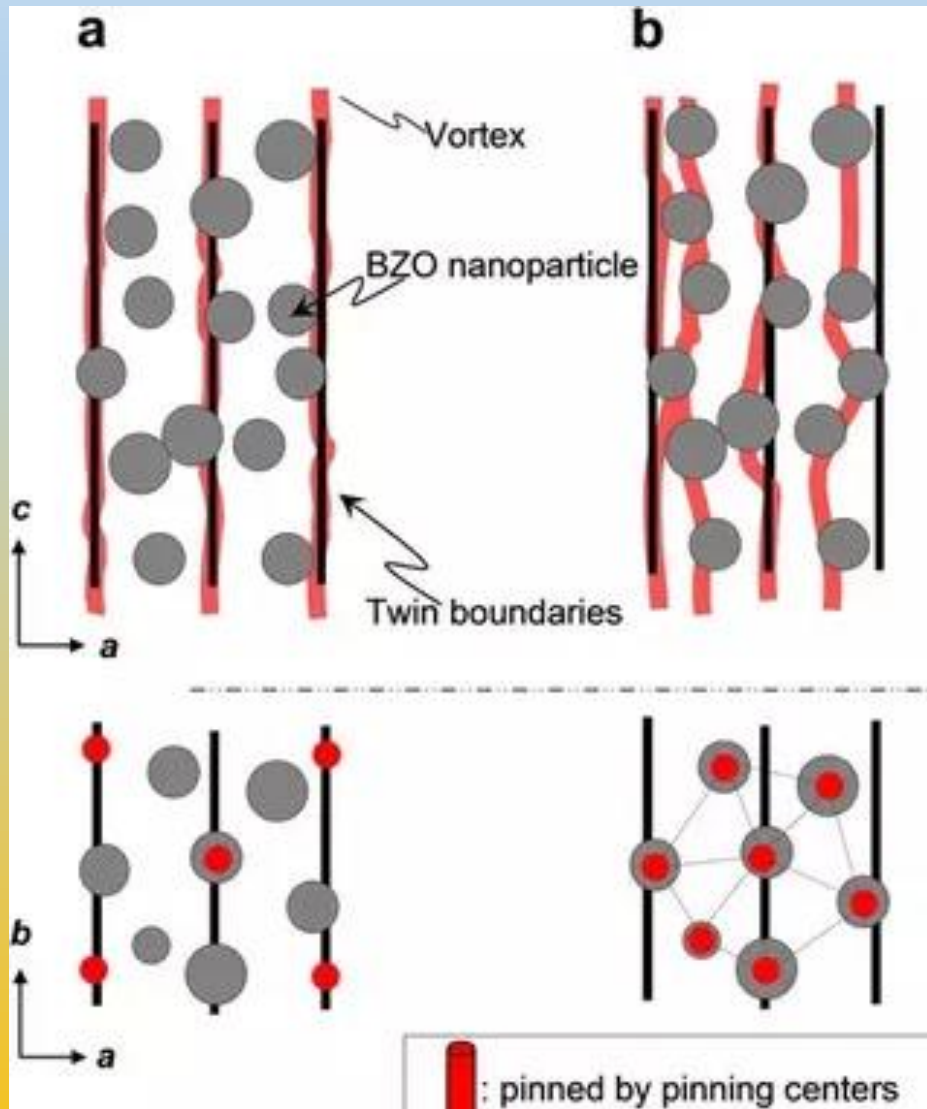
- ⇒ an **applied current** exerts a force on the vortices
- ⇒ the motion of the fluxons whose core is made up of **normal electrons** causes dissipation
- ⇒ The system is superconducting but it **is resistive and dissipates**



Vortex pinning

The flux motion reduces the current that the superconductor can carry without dissipation.

SOLUTION: to introduce defects which pin the vortices



Critical current I_c :
the maximum current that
the superconductors can
carry without dissipation

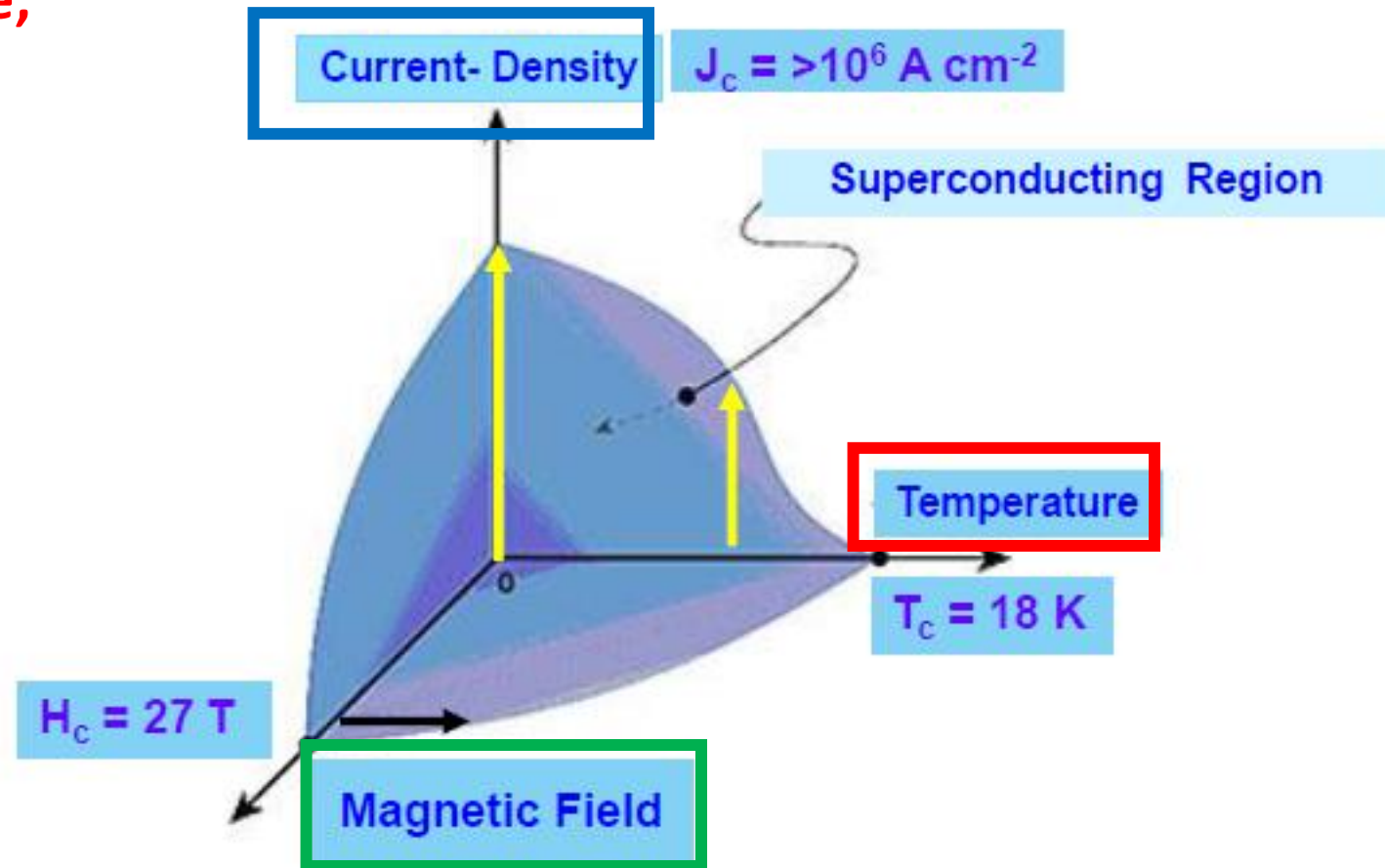
The region in which a superconductors can work without dissipation is limited by the three parameters:

Nb_3Sn (NbTi)

The critical temperature;

The critical field;

The critical current.



High Current and high Field Applications since 1961

J.E.Kunzler *et al.*, Phys.Rev.Lett. 6:3 (1961) 89

The superconducting materials

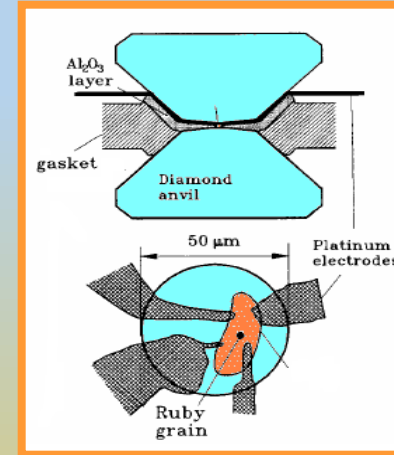
Superconductive elements

Future ?

KNOWN SUPERCONDUCTIVE ELEMENTS															
BLUE = AT AMBIENT PRESSURE GREEN = ONLY UNDER HIGH PRESSURE															
1	2													10	11
1	2													10	11
3	4													10	11
3	4													10	11
11	12													18	19
11	12													18	19
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84
87	88	89	104	105	106	107	108	109	110	111	112				
87	88	89	104	105	106	107	108	109	110	111	112				

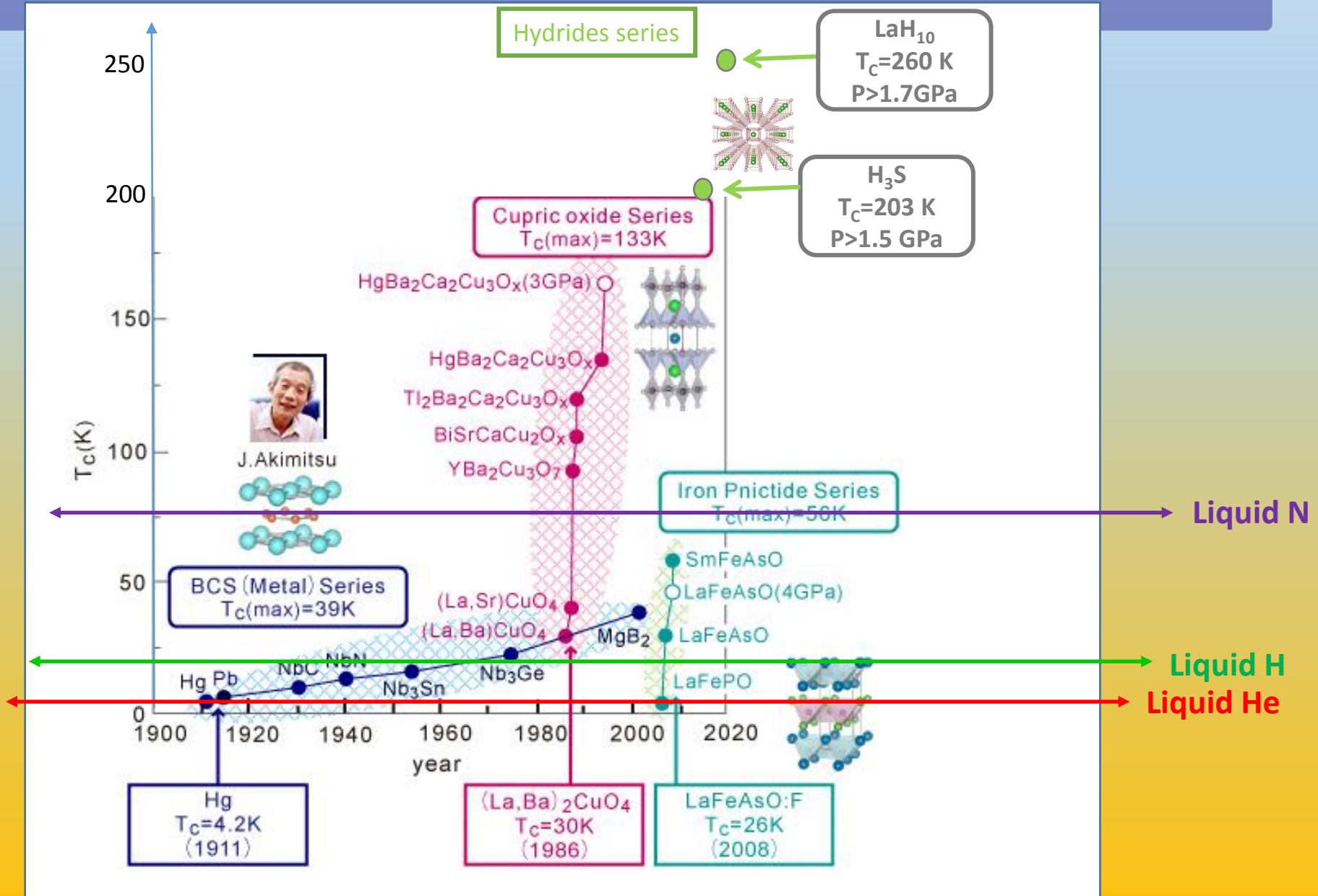
* Lanthanide Series	58	59	60	61	62	63	64	65	66	67	68	69	70	71
* Lanthanide Series	58	59	60	61	62	63	64	65	66	67	68	69	70	71
+ Actinide Series	90	91	92	93	94	95	96	97	98	99	100	101	102	103
+ Actinide Series	90	91	92	93	94	95	96	97	98	99	100	101	102	103

SUPERCONDUCTORS.ORG



Record Pressure:
250 GPa
2,500,000
atmospheres

Story of the superconducting materials

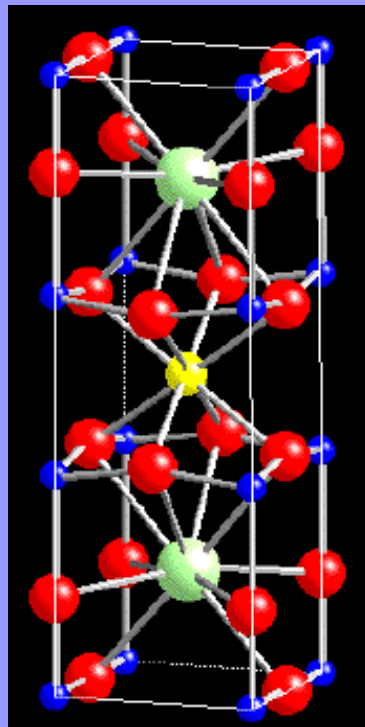


High Temperature superconductors (HTS)

1986 Bednorz and Müller 1986



1987



$\text{YBa}_2\text{Cu}_3\text{O}_7$

$T_c = 92 \text{ K}$

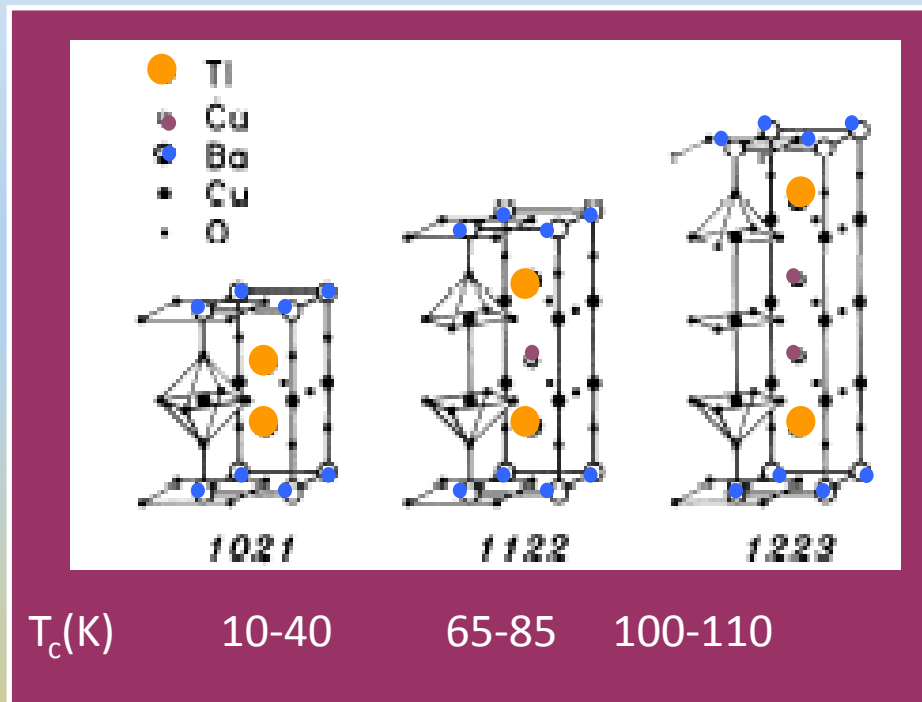
● Cu ● O

● Y ● Ba

■ Ceramic materials characterized by CuO_2 layers spaced by a reservoir block

■ Conduction (and superconductivity) occurring in the CuO_2 layers

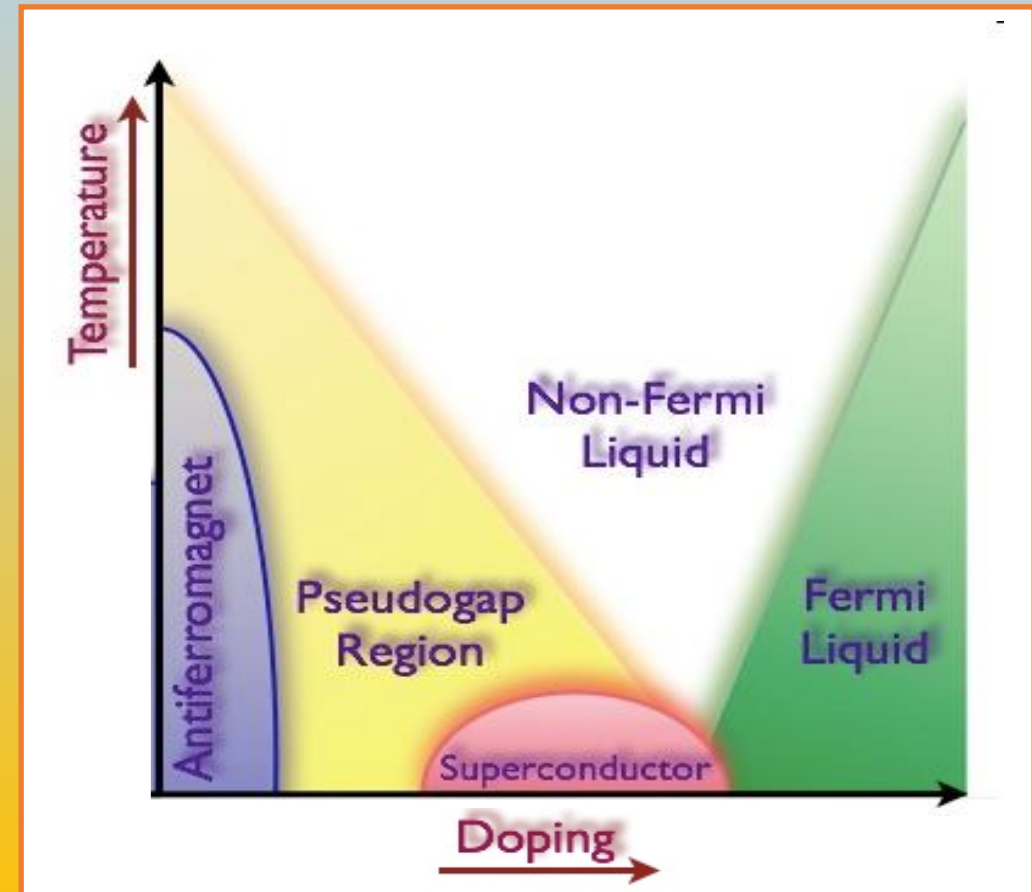
■ Maximum $T_c = 165 \text{ K}$ in $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_{8+d}$ under pressure



■ The increasing of CuO_2 layer increases T_c

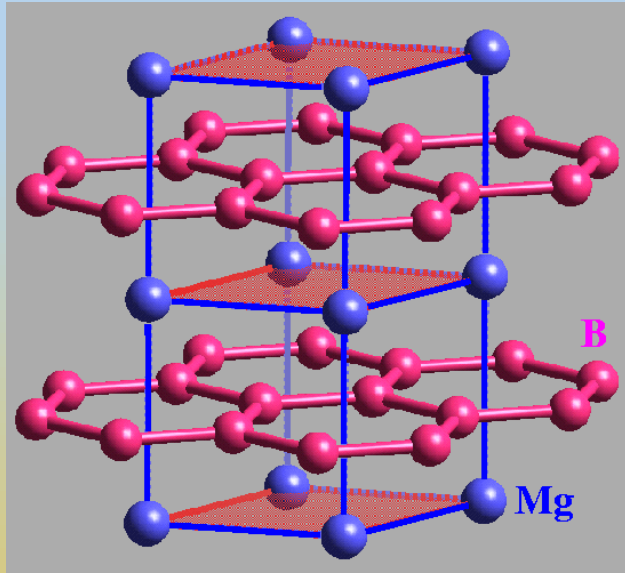
■ Complex phase diagram as a function of the doping in the CuO_2 planes

■ The coupling between electron is determined by unconventional mechanism (not mediated by the phonons) not yet completely clarified



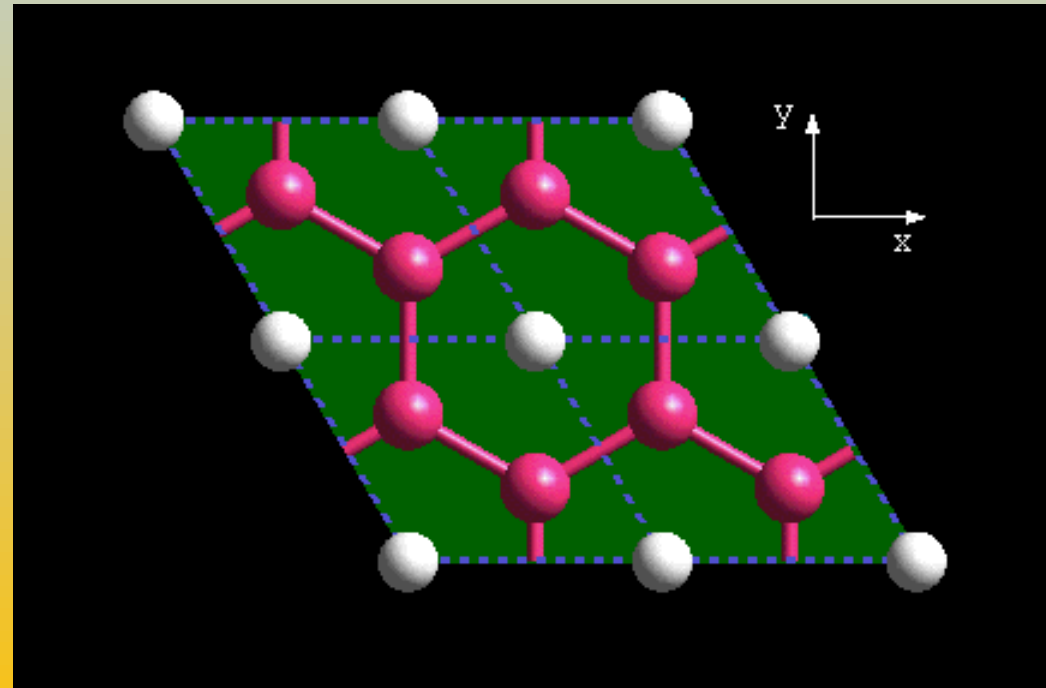
MgB₂

Akimitzu (2001)



- $T_c = 39$ K
- Layered structure
- Metallic system
- Electron pairing mediated by the optical mode of B ions
- Conventional superconductivity

$T_c = 39$ K is too near the maximum T_c predicted by the BCS theory!!



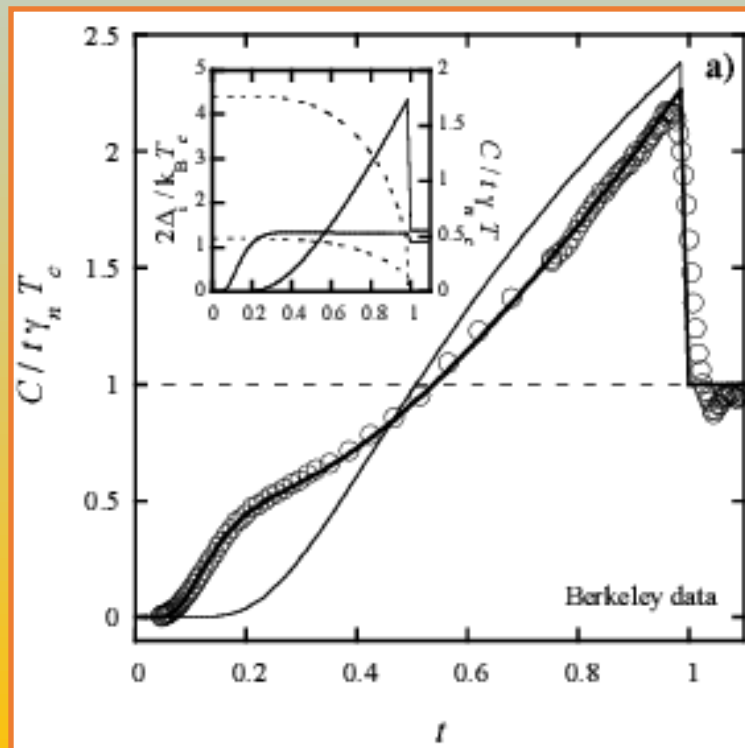
MgB₂ first example of a two-gap superconductor

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PHYSICAL REVIEW LETTERS

23 JULY 2001

Specific Heat of Mg¹¹B₂: Evidence for a Second Energy Gap

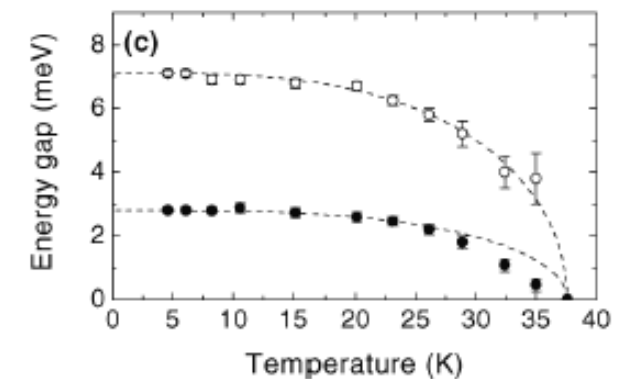
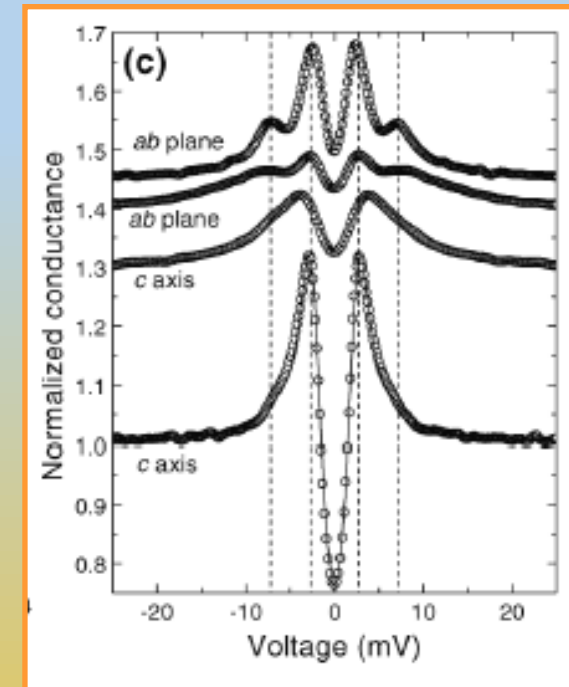


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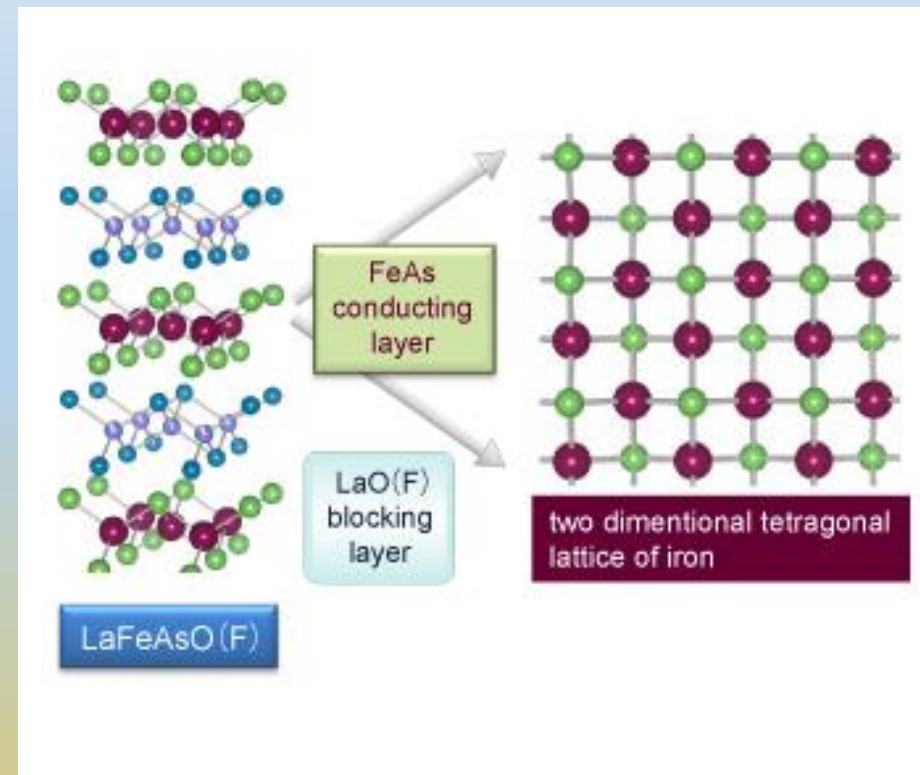
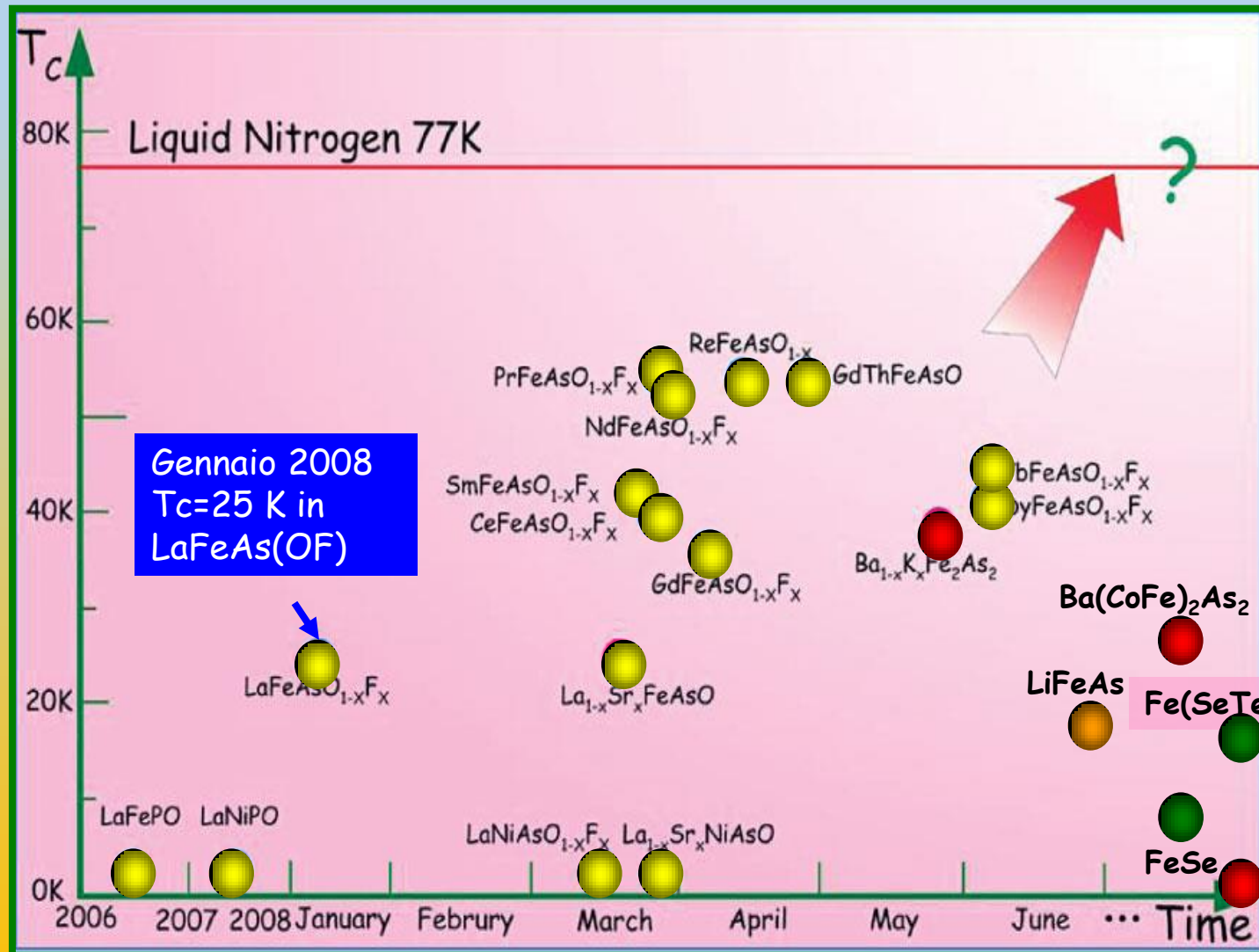
PHYSICAL REVIEW LETTERS

9 DECEMBER 2002

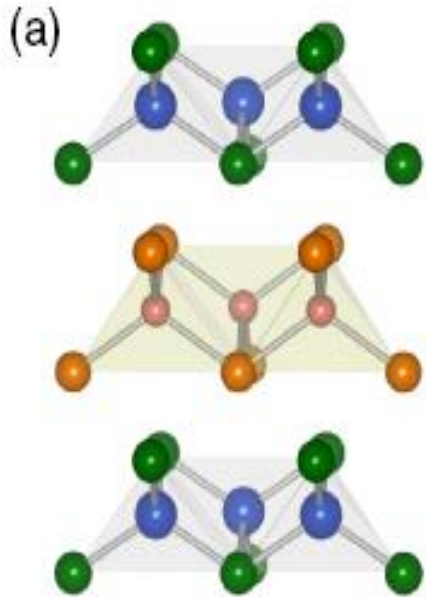
Direct Evidence for Two-Band Superconductivity in MgB₂ Single Crystals from Directional Point-Contact Spectroscopy in Magnetic Fields



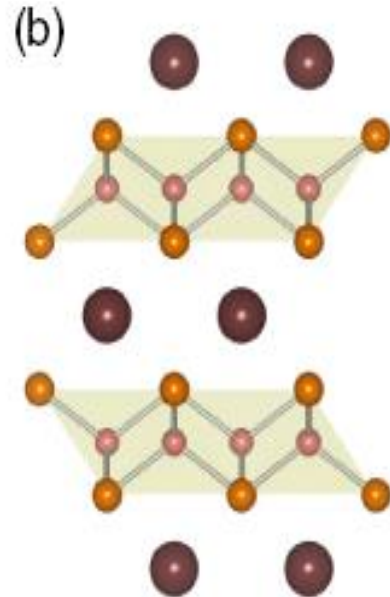
Iron based superconductors



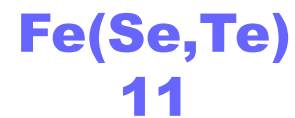
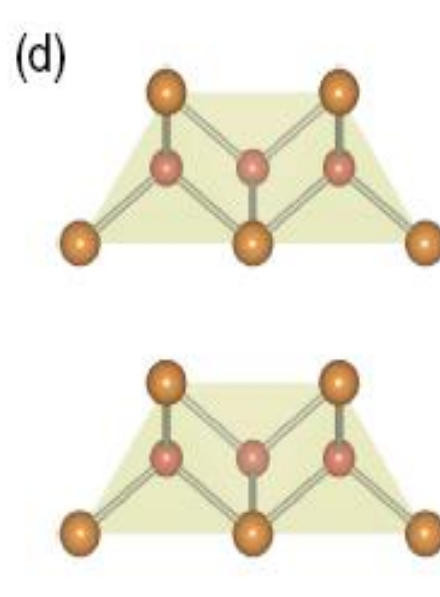
The most famous families



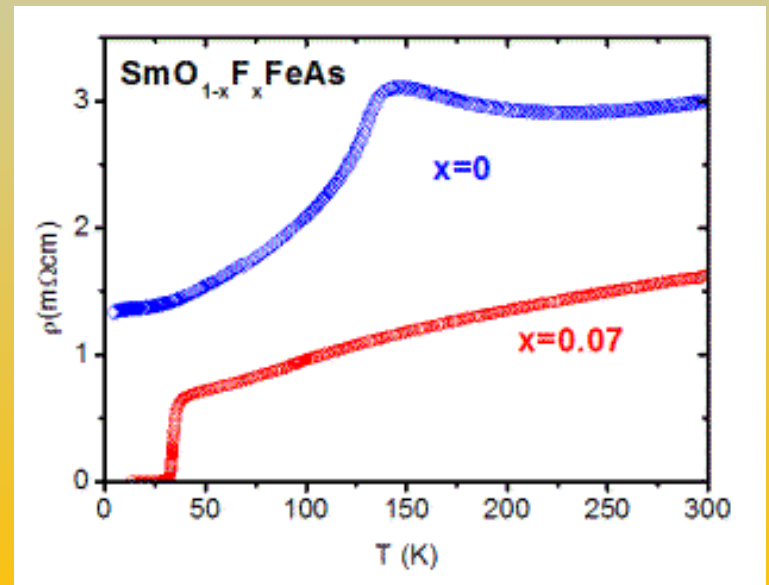
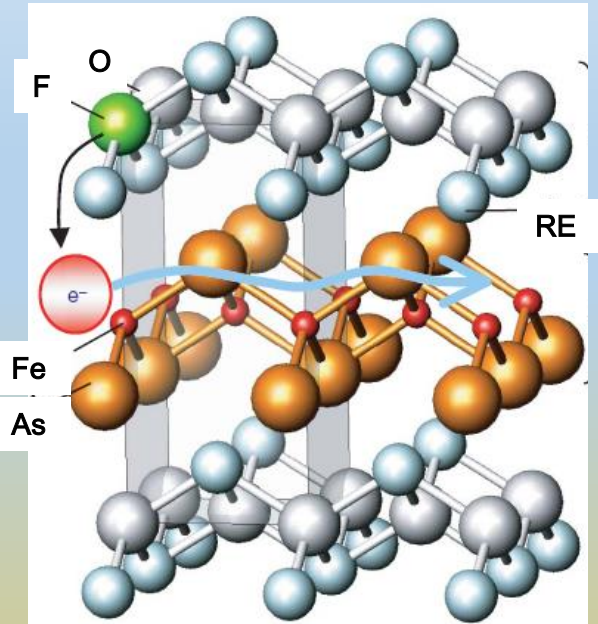
$$T_{c,max} = 58 \text{ K}$$



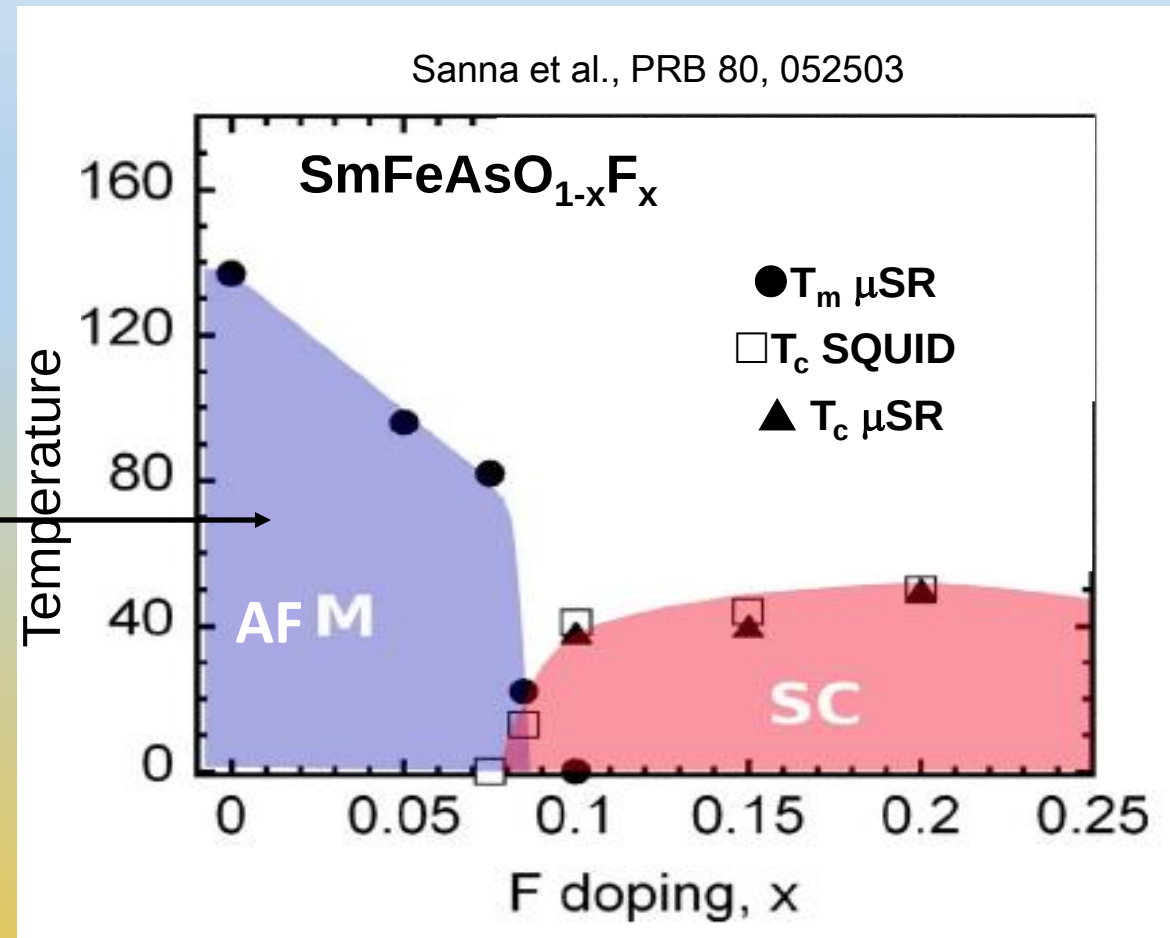
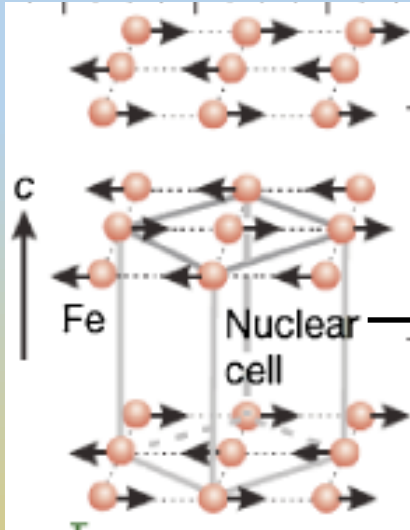
$$T_{c,max} = 40 \text{ K}$$



$$T_{c,max} = 21 \text{ K}$$



Phase diagram as a function of doping



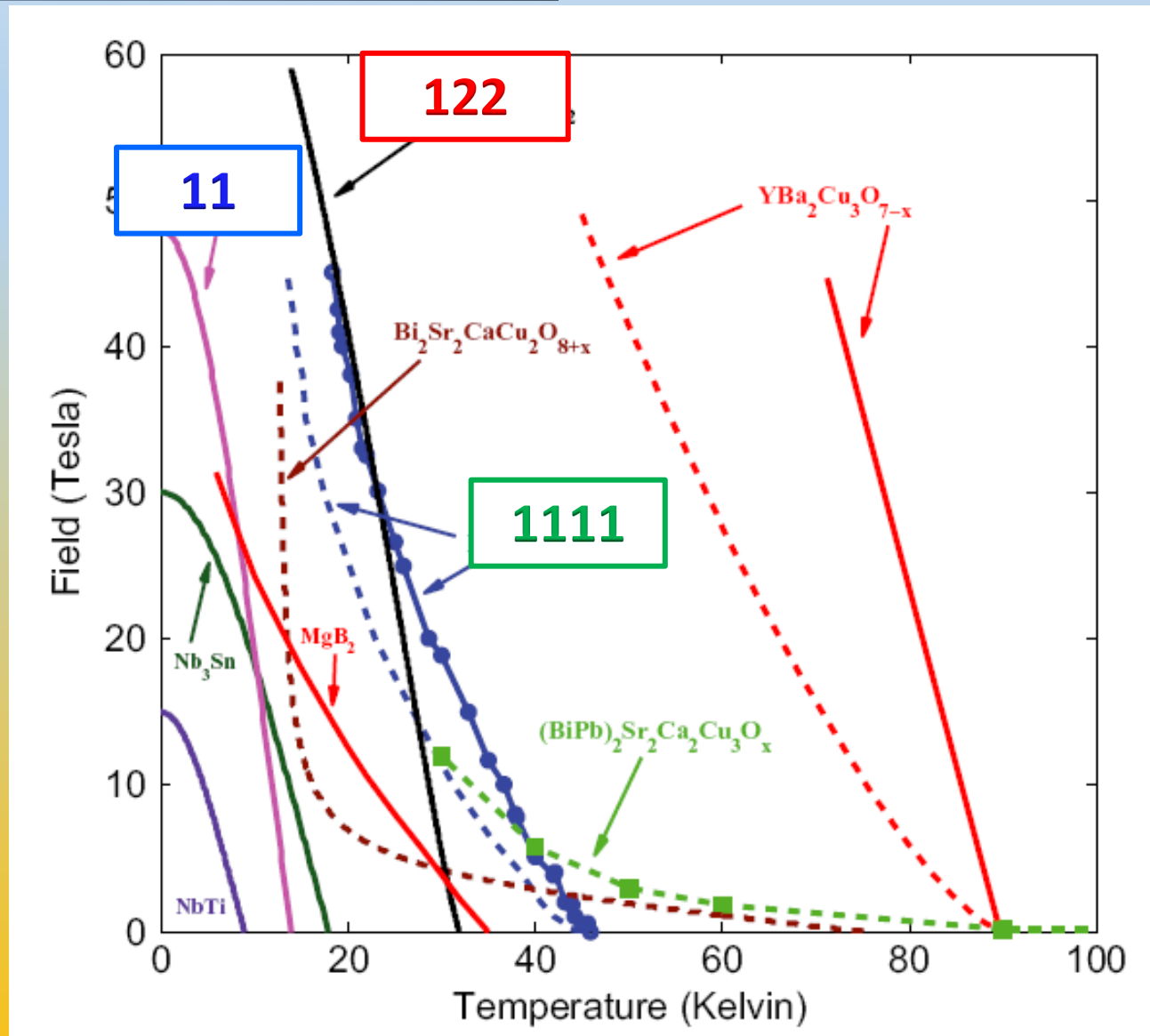
**Analogies with
HTS**

and with MgB_2

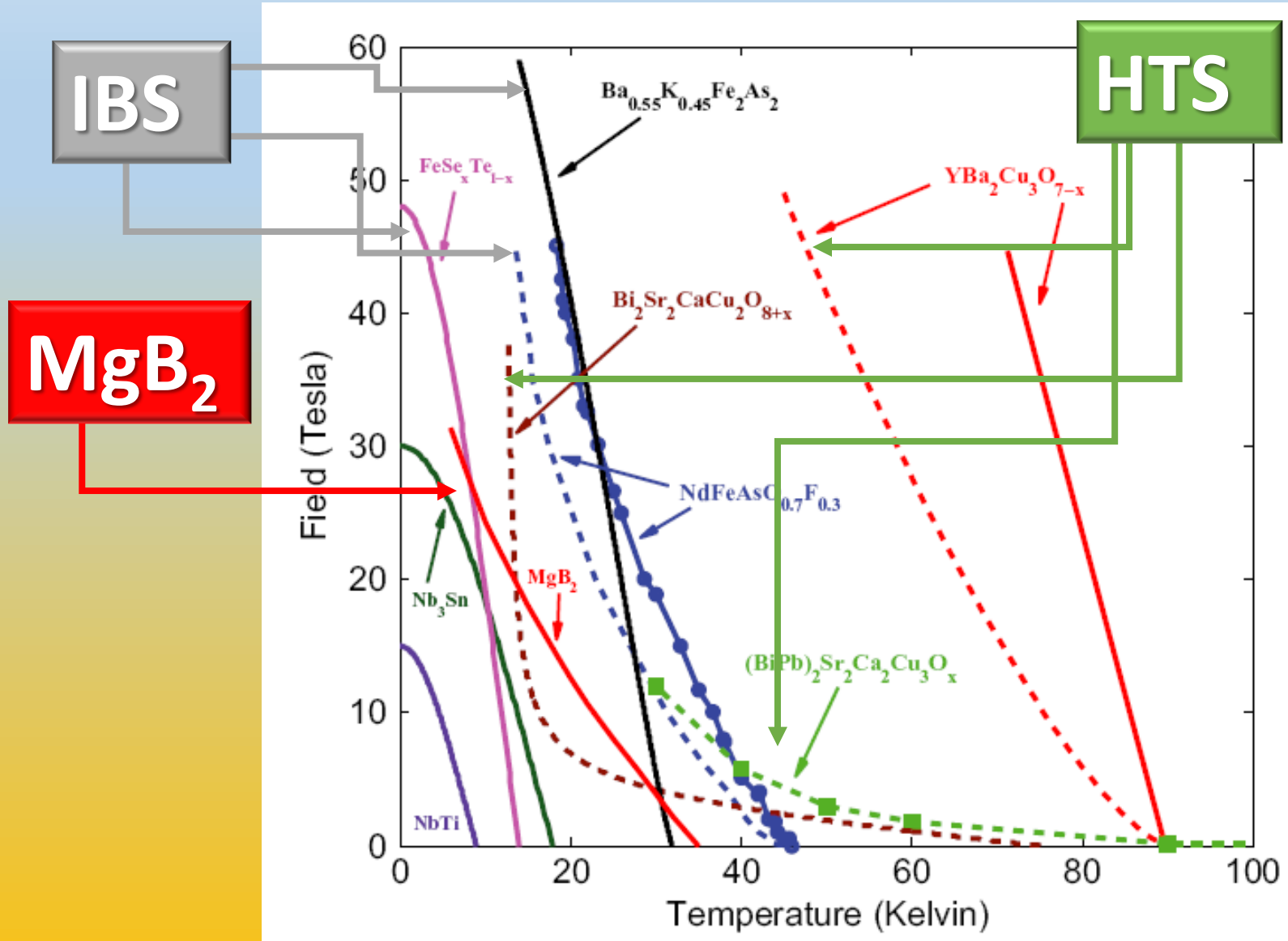
- superconductivity occurs upon doping
- proximity with magnetic ordering
- Anisotropic structure

- two-gap superconductivity

Upper critical Fields



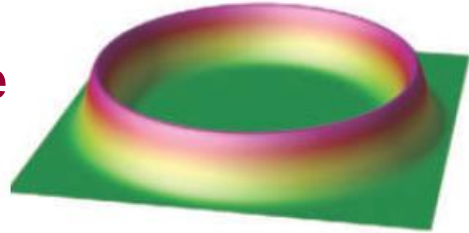
Upper critical Fields



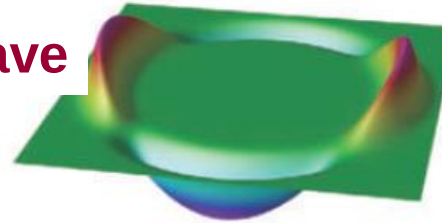
Peculiarities and issues of novel superconducting materials

1: Not conventional order parameter (Cooper pair wave function)

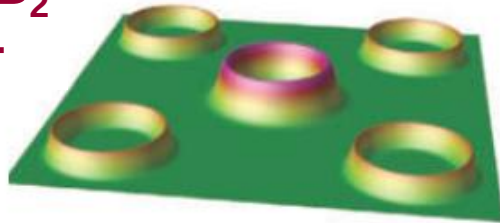
LTS
s-wave



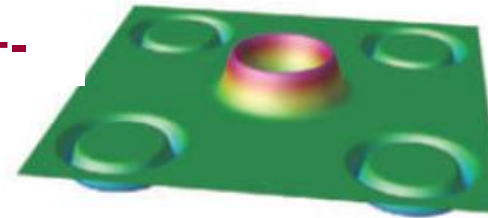
HTS
d-wave



MgB₂
s++

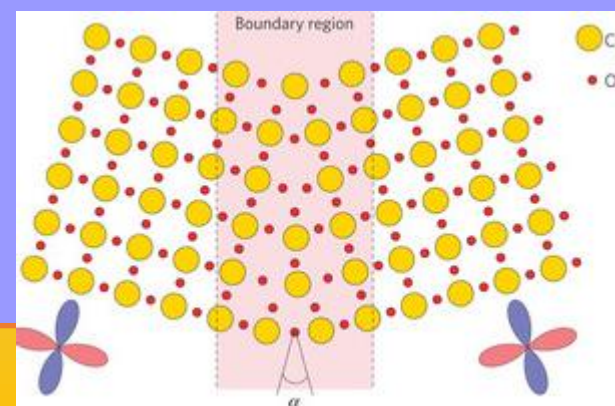


Fe-S
s+-



What does it imply
a not conventional order parameter ?

Weak coupling between grains



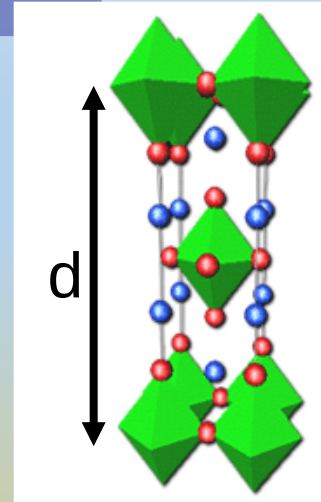
Peculiarities and issues of novel superconducting materials

2: Small coherence length

$$\xi \propto \frac{n_s^{1/2}}{T_c}$$

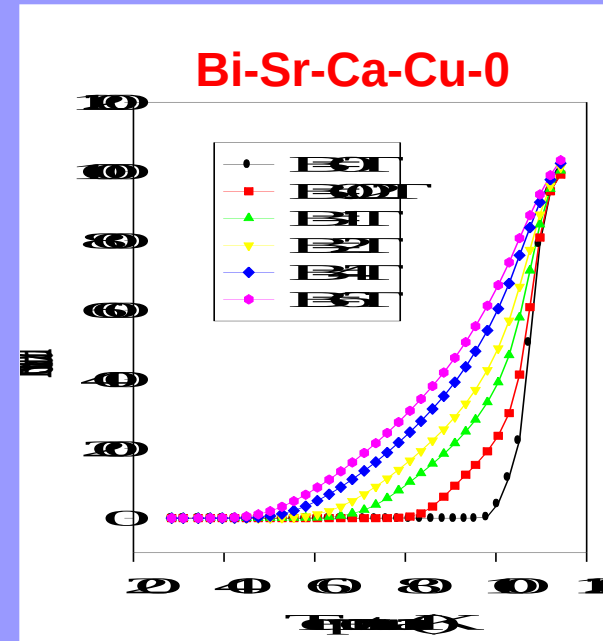
3: Large structural anisotropy

$$\xi_c < \frac{\xi_{ab}}{\gamma} < d$$



Small ξ
and large anisotropy:
What does it imply ?

Giant dissipation in presence
of an applied magnetic field





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